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Blockchain-driven security for IoT networks: State-of-the-art, challenges and future directions

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Abstract
The surge in adopting Internet of Things (IoT) devices has presented fresh challenges regarding security and networks. Conventional centralized methods encounter scalability issues and security vulnerabilities with the rapid growth of connected devices. This comprehensive review offers researchers, practitioners, and policymakers valuable insights into the current state, challenges, and future directions of blockchain-driven security for IoT networks. The paper reviews research papers from 2016 to 2023 from the leading databases. Despite the potential advantages, integrating blockchain into IoT security presents its own set of challenges. This paper also discusses a survey of surveys and explains the challenges of offering security solutions for IoT networks. The paper introduces a secure system for sharing healthcare data, incorporating AI, smart contracts, and the InterPlanetary File System (IPFS). The initial proposed system establishes a secure and transparent system for device authentication, data integrity, and access control using BC's decentralized and unchangeable characteristics. Smart contracts are employed to automate device network processes, ensuring tamper-proof execution and reducing the risk of unauthorized access. By conducting a case study and experimental assessment, we illustrate how security solutions driven by BC prove effective in mitigating the weaknesses linked to conventional IoT networks. The findings indicate that the proposed framework enhances security and provides a scalable and decentralized foundation for managing the ever-expanding ecosystem of IoT devices. The paper provides valuable guidance and outlines potential directions for future research in blockchain-driven security for IoT networks, offering a clear roadmap for researchers.

Keywords Internet of things (IoT) · Blockchain (BC) · IoT attacks · IoT device network · IoT device security

1 Introduction

Internet of Things (IoT) stands out as one of the most promising and captivating technologies of recent times, captivating academia and industry alike. Various architectural frameworks seamlessly merge the physical world with the Internet, utilizing existing network infrastructure. The primary goal of IoT is to interconnect intelligent devices, encompassing physical devices, vehicles, and home appliances, empowering them to gather and exchange data autonomously via the Internet. The concept of the IoT has transformed from a basic idea into one of the most influential drivers of business development since its introduction by Kevin Ashton in 1999 [1]. IoT is foundational for advancing data-driven digital services combined with cloud computing, big data, and machine learning. Examples of IoT networks now range from wearable devices to hardware development platforms.In 2019, there were around 8.6 billion Internet-connected devices globally. Fast forward to 2023, and the number of IoT-connected devices has surged to 15.14 billion. Projections indicate a further increase to 29.42 billion by 2030 [2], as illustrated in Fig. 1(a). This growth signifies IoT's expanding influence and ubiquity in our interconnected digital landscape. [3] The number of Active IoT devices by region is shown in Fig. 1(b). The graph is witnessing an extraordinary surge in 2023, driving the convergence of the

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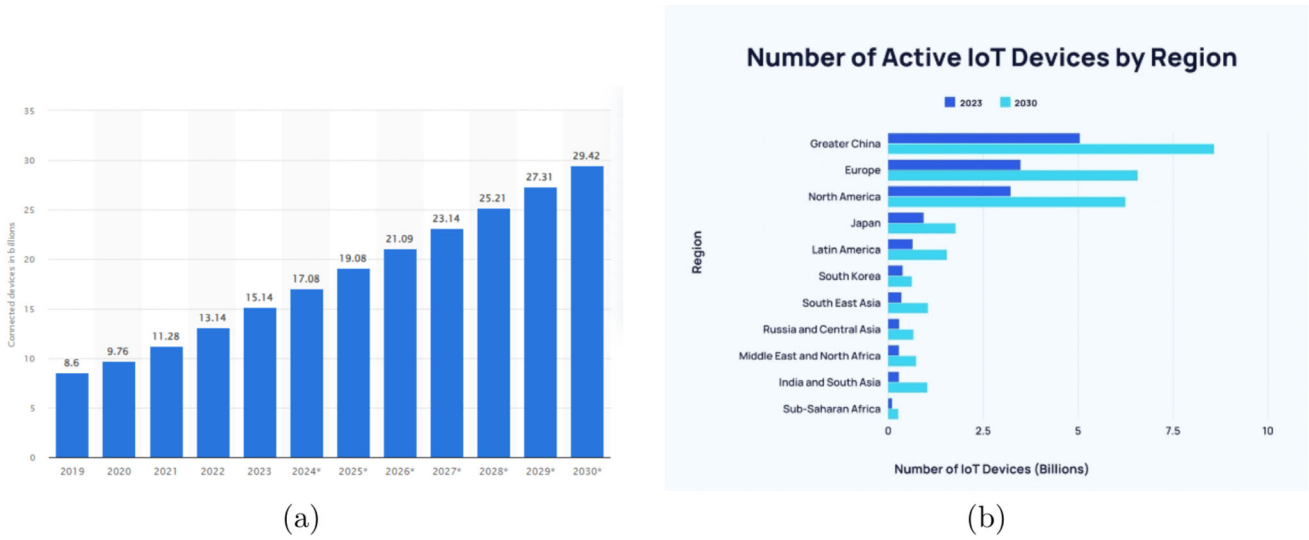


Fig. 1 (a) Number of Connected IoT devices world-wide [2] (b) Number of Active IoT devices by Regions. [3]

physical and digital realms. Projections indicate an unprecedented IoT market, estimated at 98.06 billion USD, with an anticipated Compound Annual Growth Rate (CAGR) of 23.25%, reaching USD 336.64 billion within the next five years [3]. In IoT networks, devices autonomously communicate and exchange data without human intervention, owing to the ecosystem's autonomy and ubiquity. Amidst the vast landscape of billions of IoT devices and the consequential data volumes they generate, a central challenge emerges in effectively managing the data of IoT devices within IoT networks or assigning digital identities. Addressing the security of these devices against diverse threats and attacks, alongside ensuring the integrity of the collected data from these IoT devices, poses critical challenges that warrant attention [4]. Significantly, IoT presents an opportunity to accumulate substantial data, facilitating comprehensive process monitoring to detect anomalies [5] and respond effectively to failures or cyber-attacks. Therefore, delving into the primary issues within today's IoT device network reveals that

1. Weak authentication and authorization mechanisms can lead to unauthorized access and security breaches.
2. Today's IoT device network is centralized, as the increasing number of connected devices presents a larger attack surface, making IoT ecosystems vulnerable to various cyber security threats, including malware, ransomware, and unauthorized access.
3. As IoT networks grow, managing IoT devices networks becomes challenging, requiring scalable provisioning, monitoring, and maintenance solutions.
4. Many IoT devices have resource constraints, affecting their ability to implement advanced security measures and handle complex device network tasks.
5. A surge in connected devices can lead to network congestion, affecting data transmission and overall system performance.

Blockchain (BC) is a foundational concept based on a decentralized and distributed ledger that functions without the need for a central authority, such as an agency, bank, company, or organization. It enables users to perform tamper-resistant transactions on a distributed network, with transactions recorded as a chain of blocks continuously growing with each new addition. This feature makes BC a compelling technology for developers and researchers in the IoT domain, allowing for precise recording and tracking of every transaction or data sample from their respective devices. Numerous IoT devices generate and exchange substantial amounts of private and critical data [6]. Given their limited storage, processing, and network capabilities, these IoT devices are more vulnerable to attacks than other endpoint devices like tablets, smartphones, or computers [7]. Traditional security methods [8] often result in noisy or incomplete data, potentially impairing specific IoT applications and necessitating the implementation of appropriate security mechanisms. Therefore, IoT requires scalable, lightweight, distributed security and privacy protection solutions. A secure encryption technique ensures data confidentiality as IoT data traverses multiple hops within a network. BC technology can address these challenges due to its immutable, distributed, and secure nature [9]. The significance of data in the IoT domain cannot be overstated. It is a crucial asset, utilized in various sectors such as power plants, healthcare, transportation, and social media. As computer-aided human functions increasingly rely on data, trust has become essential. However, the critical role of data has also made it an attractive target for attackers seek-

ing to compromise key qualities such as privacy, coherence, and accessibility. This underscores the need for robust security mechanisms in the IoT ecosystem, which can be provided by Blockchain technology.

1.1 Comparison with the existing surveys

Several devices, such as Smartwatches, Smart fire alarms, and Smart door locks, stand out as extensively discussed BC applications for securing IoT devices. Blockchain technology emerges as the sole technology addressing the security of IoT devices, prompting investigations into its adoption in various studies. The adoption of Blockchain technology in IoT devices has been a recurring investigation theme, yielding numerous published reviews that provide an overview of the current situation and propose research paths forward. Motivated by this, various IoT devices have been systematically reviewed in the proposed survey. This section compares state-of-the-art works, focusing on Blockchain and IoT device network and their integration. (Table 1) presents a relative comparison between existing and proposed surveys, highlighting the novelty and relevance of our research.

1.2 Motivation and contributions

1.2.1 Motivation

The IoT industry uses a traditional centralized client/server model that requires a network administrator to govern the network. Regarding IoT security, this centralized authority needs to be improved. IoT devices and sensors rely on this authority to determine how they act to function normally. If the central authority's security is breached, the information supplied by smart devices is primarily at the mercy of hackers because vast amounts of data may be obtained in one go; cyber hackers find targeting the central authority particularly enticing. As the IoT ecosystem continues to burgeon, encompassing an ever-growing multitude of interconnected devices, the traditional security paradigms face unprecedented challenges. The inherent vulnerabilities within centralized systems and the escalating sophistication of cyber threats demand innovative solutions. BC technology emerges as a potent enabler, offering decentralized, transparent, and immutable characteristics that have the potential to revolutionize security frameworks within IoT networks. The motivation lies in harnessing the unique attributes of BC, such as its cryptographic security and tamper-resistant distributed ledger, to provide robust defences against unauthorized access, data breaches, and other cyber threats. By exploring the synergy between BC and IoT, the aim is to pave the way for resilient, trust-centric security solutions that can safeguard the integrity, privacy, and functionality of IoT networks in the face of an evolving threat landscape. This exploration seeks not only to mitigate current security challenges but also to contribute to the establishment of a secure and scalable foundation for the future evolution of IoT networks.

1.2.2 Contributions

The current state of BC technology implementation in IoT networks is investigated in this survey. The study involves a comprehensive examination of the application of BC in IoT device networks. The survey contributes significantly to the field, including theoretical advancements related to the utilization of BC in IoT device networks. Significant Contributions of this survey are as follows:

1. The paper examines solutions driven by Blockchain for IoT networks, addressing security gaps/challenges through a unique survey of surveys and a literature review of existing research from the year 2016 to 2023.
2. A case study on Smart Healthcare IoT Systems has been presented.
3. The paper also introduces a secure system for sharing healthcare data, incorporating AI, smart contracts, and the InterPlanetary File System (IPFS)
4. Future research directions in security and Task allocation in IoT device networks have been identified and discussed.

The author's contribution to the survey's execution is a comprehensive understanding of how BC technology is being implemented and adopted in the security of IoT device networks.

1.3 Methods and materials

For the proposed survey, a systematic analysis was conducted to comprehensively explore BC's IoT integration. The IoT security literature is initially examined through reputable sources, including Springer, Science Direct, ACM, Taylor Francis, Wiley, IEEE Explore, and Google Scholar. Our focus shifted to BC-based system applications, investigating their potential for enhancing security in IoT device networks. To narrow our scope, we used search criteria involving BC Technology, IoT devices, BC Usage in IoT device security, and network of IoT devices. Subsequently, we extracted relevant methods from the conclusions and abstracts of the collected publications. This process aided in identifying pertinent approaches and suggestions that formed the basis for the presented survey.

Table 1 Relative Comparison between Existing Surveys and Proposed Survey

Refer ence	Year	Insights	Method Used	Limitation	Contributions
Hassan et al. [10]	2019	Privacy issues in Blockchain-IoT and five preservation strategies.	Anonymization, encryption, private contracts, mixing, differential privacy.	Privacy challenges in Blockchain-based IoT systems.	Privacy strategies and future research directions.
Viriyas itavat et al. [11]	2019	Using Blockchain in IoT for secure decentralized architecture.	Characteristics of IoT and Blockchain, business opportunities.	Lack of inherent security in IoT applications.	Secure decentralized architecture and business opportunities.
Mohsin et al. [12]	2019	Taxonomy of BC in networking authentication.	Comprehensive taxonomy and categorization.	Limited source databases, outdated articles.	Detailed taxonomy for BC in networking authentication.
Cui et al. [13]	2019	Blockchain benefits, challenges, and roadmap for IoT.	Detailed taxonomy and applications.	Heterogeneity and scalability issues.	Comprehensive analysis and taxonomy for BC in IoT.
Lo et al. [14]	2019	Literature review on Blockchain-IoT integration.	Systematic literature review, evaluation methods.	N/A	Summary of IoT issues and Blockchain solutions.
Mistry et al. [15]	2020	Review of BC in 5G-enabled IoT for industrial automation.	Survey and comparison of proposals.	Centralized architecture issues, security concerns.	Comprehensive survey of BC-based industrial automation.
Rao and Clark [16]	2020	Blockchain applications and challenges in IoT devices.	Survey of applications and strategies to address challenges.	Trust, security, efficiency challenges, cybersecurity implications.	Survey of promising Blockchain applications in IoT.
Uddin et al. [17]	2021	Advances in Blockchain for Cloud and Fog IoT.	Analysis of advancements, obstacles, solutions.	Privacy concerns, data flow bottlenecks, firmware issues.	Analysis of Blockchain advances and solutions in IoT.
Azbeg et al. [18]	2021	Review of healthcare applications integrating IoT and Blockchain.	Review study, database searches.	Challenges in integrating Blockchain with healthcare IoT.	Taxonomic review of IoT-Blockchain in healthcare.
Saxena et al. [19]	2021	Integration trends and security enhancements in Blockchain-IoT.	Survey on security enhancements and integration trends.	Security vulnerabilities, privacy risks.	Survey on security enhancements in Blockchain-IoT.
Singh et al. [20]	2021	Security attacks and solutions in Blockchain-IoT systems.	Two-level Blockchain mechanisms, countermeasures.	Unclear terminology, adoption risks.	Overview of Blockchain security parameters and solutions.
K. Hameed et al. [21]	2021	Decentralized BC-based authentication for IoT using tokens.	Token mechanism, high-level Petri nets, Z3 SMT solver.	Insufficient conventional approaches, extra overheads.	Efficient decentralized authentication scheme for IoT.
E. A. Shammar et al. [22]	2021	IoT and Blockchain integration challenges: clock synchronization and data security.	Network time protocol, lightweight cryptographic approach.	Communication issues, data integrity and security.	IoT-Blockchain integration with lightweight cryptography.
R. Kumar et al. [23]	2022	Deep learning and blockchain for secure zero-touch networks.	Deep learning models, blockchain framework.	High computational overhead, scalability issues.	Combines deep learning and blockchain for secure network management.
K. Randhir et al. [24]	2022	Decentralized medical image storage using IPFS, blockchain.	IPFS for storage, Blockchain for sharing.	Integration complexity, scalability issues.	Decentralized image sharing, immutability, privacy in healthcare.
A. Al Sadawi et al. [25]	2022	Integration of BC with IoT and role of oracles.	Literature-based methodology, surveys, oracles.	Incomplete view of IoT-BC integration.	Narrative on BC-IoT integration and oracles.
G. Zhang et al. [26]	2022	Blockchain-based system for securing electronic health records (EHRs).	Pairing-based cryptography, smart contracts.	N/A	Privacy-preserving e-health system with tamper-proof records.
A. A. Khan et al. [27]	2022	BC framework for IoT security in industrial domains.	Literature review, proposed framework.	Performance issues, high resource requirements.	Secure framework for industrial IoT.

Table 1 continued

Refer ence	Year	Insights	Method Used	Limitation	Contributions
S. Ait Bennacer et al. [28]	2022	Private Blockchain-based digital health passport for COVID-19 in Morocco.	On-chain and off-chain storage (IPFS).	Unauthorized access, data protection compliance.	Digital health passport for privacy and security.
P. Kumar et al. [29]	2022	AI-based intrusion detection at the edge enhances security. Blockchain ensures data integrity and decentralized trust	AI for initial data filtering. Blockchain with smart contracts for transaction validation. IPFS for efficient data storage	High complexity and cost. Performance issues like low transaction throughput and high latency. Increased energy consumption and costs.	Secure and efficient data dissemination in IoTEHS. Enhanced data security and privacy.
K. Randhir et al. [30]	2022	Decentralized storage model removes third-party dependency	Blockchain for secure, immutable data sharing	Potential scalability issues with large datasets	Enhances data privacy and accessibility in healthcare systems.
C. Wang et al. [31]	2023	Sustainable architecture for digital twins using Blockchain for IoT.	Sustainable data collection and BC-based architecture.	Inefficient networks and scarce resources.	Sustainable BC-based architecture for digital twins.
S. Basudan et al. [32]	2023	Fuzzy Blockchain framework for threat detection in IoT networks.	Fuzzy DL model, meta-heuristic algorithms.	Inadequate for addressing threats solely with BC.	Secure fuzzy BC framework for attack detection.
C.hang Cheng et al. [33]	2023	Blockchain-based access control for IoT networks.	ABAC system using Ethereum.	Auditability, resource consumption, scalability issues.	ABAC access control system for IoT.
V.A.R Iboomsom et al. [34]	2023	Blockchain-enabled IoT in healthcare for data security, privacy, decentralization.	PRISMA review, Scopus database.	Centralized processing risks data manipulation.	Highlights Blockchain's role in smart contracts, patient care, supply chains.
C.Trivedi et al. [35]	2023	Shift to decentralized IoT ecosystems for security, privacy.	Low-powered consensus, smart contracts.	Centralized IoT security, resource constraints.	Case study on blockchain-driven IoT ecosystem.
V. Gugueoth et al. [36]	2023	Security, privacy threats in IoT, addressed by blockchain.	Systematic review, PRISMA, Scopus data.	Centralized processing risks, resource constraints.	Analyzes consensus protocols, security techniques, suggests research directions.
H. Chen et al. [37]	2023	Hybrid IoT architecture using blockchain for security, privacy.	Fog, edge computing, software vulnerability analysis.	Centralized processing, longer on-chain time.	Proposes blockchain-as-a-service, addresses seven challenges.
R. Kumar et al. [38]	2023	Blockchain and XAI for secure consumer IoT, focusing on transparency.	Blockchain authentication, LIME XAI.	Centralized processing risks.	Framework for security, interpretability in IoT with a case study.
P. Kumar et al. [39]	2023	XAI and Blockchain in CIoT-inspired Metaverse for security, privacy.	Systematic review, use case demonstration.	Centralized processing risks.	Proposes XAI and Blockchain architecture for Metaverse security.
P. Kumar et al. [40]	2023	Blockchain and AI in IoT e-Health for security, privacy.	AI-based intrusion detection, Blockchain, IPFS.	Centralized processing risks.	Secure data framework for IoT e-Health, resists various attacks.
The proposed survey	2024	BC application for IoT security: access control, data, and identity management.	Smart contract and ML for attack detection.	Regulatory compliances	Blockchain and ML-based approach for IoT security.

1.3.1 Research questions

The research questions, which represent the overall goal of the work, heavily influence the selection of primary studies. Understanding and evaluating BC technology and looking into whether it can effectively address the problems with current IoT device network systems are the main objectives of this research. We looked into the potential of BC in secure IoT devices due to several study-related issues. They are listed below:

1. RQ1: What novel approaches can be developed to ensure efficient and resource-friendly data synchronization and validation in IoT device networks?
2. RQ2: Is BC a viable solution for addressing IoT's technical and security challenges?
3. RQ3: What are the Impact areas of BC Utilization in IoT device network?
4. RQ4: Identify the possible weaknesses and attack vectors unique to BC-integrated IoT networks.

1.3.2 Data sources

To obtain the most recent literary studies and online material, we looked through standard reviewed journal databases, including IEEE Xplore, Elsevier, Science Direct, and Springer. We also used other sources, such as case-specific technical books, concessions, websites for imagining, and online publications associated with the current study.

1.3.3 Search strings

Certain keywords and their synonyms are chosen from the identified research topics to create the search string. Then, in a specific order, the keywords are arranged using the conditions "AND" and "OR" to create the following search: *(("BC" OR "BC Technology" OR "IoT" OR "IoT Security") AND ("IIoT" OR "IIoT Security" OR "IoT device and their security") AND ("IoT device network " OR "IoT" OR "Internet of Things network")).*

1.3.4 Criteria for inclusion

The requirements for inclusion are listed below.

1. Papers released between 2016 and the end of 2023.
2. Publications presented at or published in proceedings from conferences, peer-reviewed journals, or books.
3. Publications highlight the need for secure IoT devices.
4. This paper focuses on implementing BC Technology in the Internet of IoT devices, particularly on recent publications.

1.3.5 Criteria for exclusion

The criteria for exclusion are listed below.

1. Publications released before 2016.
2. Documents written in a language other than English.
3. Papers that do not specifically address the use of BC technology in IoT networks are also rejected.

Figure 2 depicts that 286 papers were retrieved from the database until the end of 2023. After a rigorous review, 149 papers considered essential to our review were shortlisted and thoroughly examined.

1.3.6 Structure of this survey

The survey is structured as follows: Section II discusses IoT and BC technology principles. Following a literature review in Section III, the discussion centers on using BC technology in managing IoT device networks. Section IV identifies a case study on Smart healthcare IoT systems, Section V provides future scope and research direction, and Section VI provides

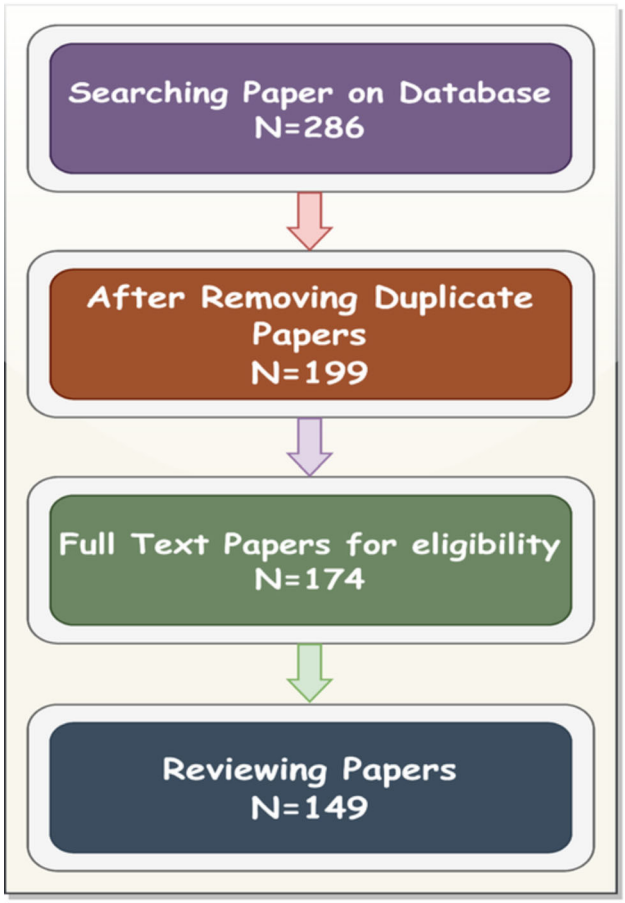


Fig. 2 Survey Criteria (N=No. of Papers)

an overall conclusion and discussion. The survey structure is illustrated in Fig. 3.

2 Internet of things and BC technology: Background, definition, and motivation

This section delves into the history of IoT and BC Technology. It covers the fundamentals of BC Technology, elucidating how it operates, and finally focuses on the conclusion and discussion of BC in IoT systems.

2.1 Internet of things

The Internet of Things (IoT) envisions a future where objects seamlessly integrate into the Internet, each uniquely identified and accessible on the network, with location and status readily available. This expanded Internet combines the digital and physical realms into a unified entity enriched with services and intelligence. A diverse array of Smart IoT devices is being developed across various IT layers, each designed for specific purposes and possessing unique characteristics. Notably, the human element, while integral, is part of this interconnected system. The widespread adoption of the Internet has become essential in everyday life, and IoT is emerging as a pioneering web innovation. Positioned as the next Industrial Revolution, IoT will transform interactions among businesses, governments, and consumers, significantly increasing the variety and quantity of connected devices.

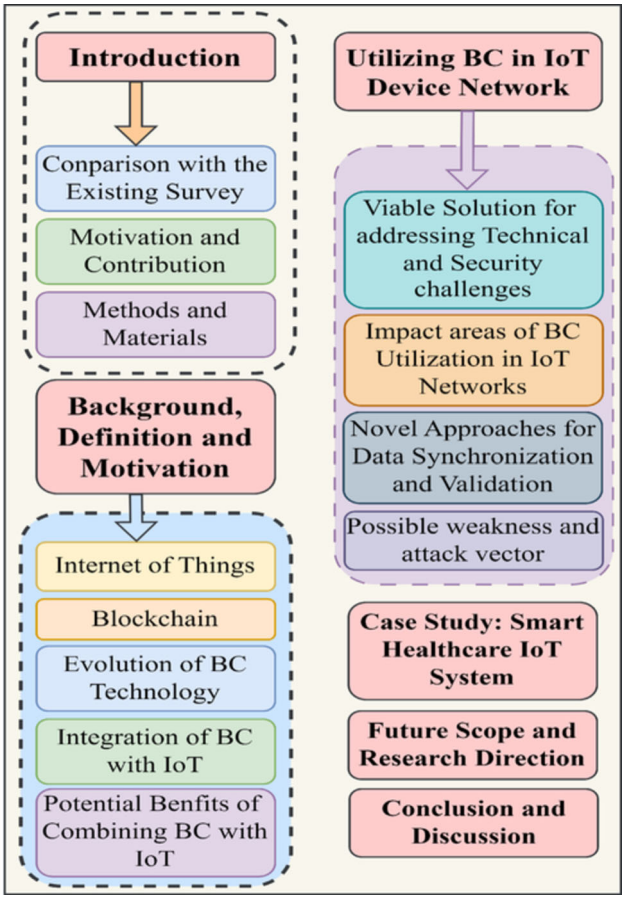


Fig. 3 Flow of this survey

2.1.1 IoT layered architecture

The Internet of Things (IoT) results from various devices and interconnected sensor networks. These devices and networks share features like small memory, low power consumption, unique identifiers, and restricted processing power [32]. Users can remotely manage data provisioning and access services through established gateways connecting the IoT environment to the outside world. Constrained by time, place, device type, or network, IoT allows seamless connectivity between people and objects [41]. The proposed designs provide different viewpoints and functionalities. Interfaces and services, network and communication, and perception and hardware are the three layers of architecture [42]. Concerning technologies such as cloud computing, fog computing, and smart computing, some experts have proposed a fourth layer, the support layer [43]. Different architectural scales and technology techniques are embodied in this fourth layer [44]. The application, data processing, network, and sensors/actuators layers are the four main components of generic IoT architectures. The application layer handles healthcare, smart homes, and transportation applications, providing results from the data processing layer [45]. This user-centric layer carries out several tasks on the user's behalf. The sensing layer's data is analyzed by the data processing layer, which then stores the results in specific IoT devices like home hubs and wearables to improve user experience. Using technologies like Bluetooth, Wi-Fi, Z-Wave, cellular networks, and Zig-Bee [23], the network layer acts as a communication channel, enabling data exchange with other connected devices. Events are detected, and real-time data is gathered via the sensors/actuators layer [45].

2.1.2 How IoT ecosystem works

Smart gadgets with embedded CPUs, sensors, and communication equipment that are made together, transmit, and respond to environmental data make up an IoT ecosystem. By connecting to an IoT gateway or edge device, devices can share sensor data and send it to the cloud or local storage for analysis. These gadgets exchange information with other connected devices, acting upon that information. The majority of jobs are completed independently, requiring less human involvement. Conversely, people can utilize these

devices for data access, setup, and instructions. Specific IoT applications depend on connectivity, networking, and communication protocols. Figure 4 shows the IoT ecosystem.

2.1.3 Characteristics of IoT

Interconnectivity, large size, safety issues, interaction, rapid modifications, and services associated with connected devices are some benefits of the Internet of Things. utilizing various hardware platforms and devices to communicate over diverse networks is known as heterogeneity. Everything is connected through a worldwide network of communication and information. Safety considers the effects on the surrounding environment, physical health, endpoint protection, and personal data. Because of the ubiquitous use of devices, the Internet of Things substantially impacts the number of endpoints linked. Network accessibility and compatibility ensure system effectiveness, involving protocol matching and managing interfaces. Accessibility refers to information access with proper authorization and authenticity anytime, anywhere. Dynamic changes refer to the evolving states of devices. Services related to things ensure semantic coherence and privacy protection within device constraints [46].

2.1.4 Challenges in IoT device network

Despite numerous advantages, IoT introduces challenges such as scalability, interoperability, security, heterogeneity, and privacy [24]. Researchers have proposed solutions to improve interoperability [47–51]. Interoperability is the capability of diverse system components to work together effectively. Scalability challenges arise from managing the rapid proliferation of devices and evaluating the system's capacity to handle future growth without compromising performance [52–55]. Heterogeneity in IoT networks is prevalent due to the variety of devices, necessitating standardized abstraction approaches and maximizing functionality across diverse configurations [24]. The proliferation of IoT has introduced new security vulnerabilities, primarily due to inherent heterogeneity, decentralization, and individual system vulnerabilities [24]. Implementing security measures in resource-limited networks is challenging, with traditional techniques like encryption, authentication, and authorization facing difficulties [48]. Autonomous approaches may be needed to address security and safety concerns effectively [49, 50]. IoT devices are vulnerable to malware, and timely security firmware updates are challenging [51]. Data privacy is another significant concern, with risks when data is uploaded to third-party cloud servers susceptible to breaches [52]. Table 2 has identified the significant challenges in IoT networks [57, 58] IoT protocols operate across different layers, exposing them to various potential attacks [19]. Possible attacks on each protocol within a specific layer include Solaris, Cross-site scripting, HTTP flooding, DDoS, and repudiation attacks targeting the application layer. The data processing layer is vulnerable to exhaustion attacks and targeted malware. Networking layer risks include injection, smurf, SYN flooding, opt-ack, Sybil, sinkhole, wormhole, and other attacks. Attacks on the BC network encompass resource consumption, Byzantine, and IP address spoofing. The sensors/actuators layer is susceptible to physical damage, access control issues, and disconnection of physical links [56]. Specific attacks based on systems like Wireless Sensor Networks (WSN) and Radio Frequency Identification

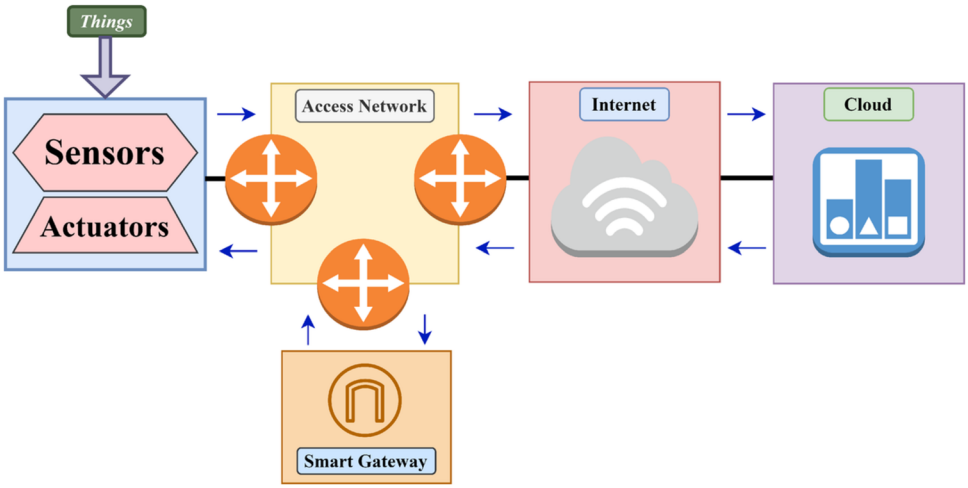


Fig. 4 Ecosystem of IoT

Table 2 Major Challenges of IoT

Challenge	Causes
Centralization	Authenticating, and identifying every device. All interconnected devices solely access the internet, regardless of their physical proximity. Employing a client/server architecture, where numerous clients are linked to a singular server.
Transparency	Data inventories that are incorrect and difficult to access. Lack of a policy ensuring the confidentiality of data.
Interoperability	The process of recognizing and verifying the identity of every device. All connected devices solely navigate the internet, regardless of their physical distance. A model where numerous clients are linked to a single server in a client/server architecture.
Security	Manipulating data from distant devices. Exploiting compromised devices. Inability to monitor all linked devices within the network. Absence of security updates for IoT devices. As the number of connected devices rises, the susceptibility of endpoints also increases.

(RFID) include synchronization issues, selective forwarding, replay attacks, sinkhole attacks, false routing, eavesdropping, buffer overflow, injection attacks, replay attacks, Sybil attacks, passive interference, impersonation, spoofing, tag modification, and buffer overflow.

a) In the context of Wireless Sensor Networks (WSN):

1. **Physical/Link layer:** Concerns include synchronization issues, selective forwarding, and replay attacks.
2. **Network/Transport layer:** Risks involve sinkhole attacks, false routing, and eavesdropping.
3. **Application layer:** Potential threats encompass buffer overflow and injection attacks.

b) For the usage of Radio Frequency Identification (RFID):

1. **Physical/Link layer:** Potential security threats encompass replay attacks, Sybil attacks, and passive interference [57].
2. **Network/Transport layer:** Risks include eavesdropping, impersonation, and spoofing.
3. **Application layer:** Security threats involve tag modification, buffer overflow, and injection attacks [58].

2.2 Blockchain technology

BC is a decentralized, digital ledger that records timestamped transactions across a series of blocks [59]. Each computer in the BC network acts as a node, collectively maintaining identical copies of the data. Nodes use the same consensus algorithm to validate transactions, enhancing network reliability by eliminating single points of failure. The blocks in a BC form a consecutive chain, linking each Block to the next. BC technology also supports smart contracts, which automatically execute agreements based on predefined conditions [60]. Implementations like Ethereum [57] and Hyperledger [61] are notable for supporting smart contracts. BC aims to provide decentralization, anonymity, autonomy, transparency, privacy, security, and collective verification [62]. Figure 5 shows the structure of the Block.

2.2.1 Consensus algorithms

Consensus algorithms establish rules and procedures to motivate nodes to demonstrate trustworthiness by generating or adding new blocks to the BC [63]. They are crucial in various sectors, including banking, finance, supply chain networks, healthcare, real estate, media, record networks, and cybersecurity. Examples include PoW [64], PoS [65], DPoS [66],

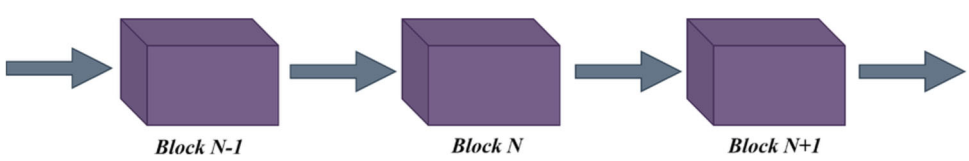


Fig. 5 Structure of Blocks

TaPoS [67], Proof of Activity [68], Proof of Capacity [69], BFT [70], PBFT [71], DBFT [72], BFTRaft, PoA [73], PoSV [74], PoET [75], and Tendermint [76]. Detailed information on these algorithms can be explored in [77–82].

2.2.2 Types of BC networks

1. **Public BC Networks:** Open and permission-less, allowing anyone to participate as a node. Examples include Bitcoin and Ethereum, which operate on PoW or PoS [83].
2. **Private BC Networks:** Permissioned networks restricting participation to trusted entities. Suitable for applications requiring privacy and control, with higher scalability and faster transactions [84].
3. **Consortium BC Networks:** Governed by a consortium of organizations, offering controlled and collaborative environments with shared governance [85].
4. **Hybrid BC Networks:** Combine public and private BC components, enabling organizations to maintain private operations while interacting with public BC for specific transactions.

2.3 Potential benefits of combining BC with IoT

BC Network has been increasing rapidly daily. Table 3 discusses the timeline highlighting the evolution of BC tech-

Table 3 Evolution of BC Technology

Years	Phases	Description
2008	Bitcoin and the BC	Satoshi Nakamoto released the whitepaper for Bitcoin, introducing the concept of a decentralized digital currency and the underlying technology called BC.
2009	Bitcoin Network launch	The Bitcoin network officially launches, allowing users to send and receive digital currency using BC technology.
2011	Alternative cryptocurrencies and BC	Introduction of alternative cryptocurrencies, such as Litecoin and Namecoin, utilize BC technology for secure transactions.
2013	Ethereum and Smart Contracts	Vitalik Buterin introduces Ethereum, a BC platform that supports the execution of smart contracts, enabling programmable decentralized applications (DApps).
2015	Permissioned BC	Introduction of permissioned BC platforms like Hyperledger Fabric and R3 Corda, which cater to enterprise needs by providing private and consortium-based networks.
2016	Consensus mechanisms and scalability solutions	The rise of alternative consensus mechanisms, such as Proof of Stake (PoS), Byzantine Fault Tolerance (BFT), and delegated consensus, seeks to address scalability and energy consumption challenges in BC development.
2017	Initial Coin Offerings (ICOs) and Tokenization	Rapid growth of Initial Coin Offerings (ICOs) as a fundraising method, allowing companies to tokenize their assets and raise capital through BC-based token sales.
2019	Interoperability and Cross-Chain Communication	Projects like Polkadot and Cosmos introduce interoperability solutions, enabling different BC networks to communicate and exchange assets.
2020	Central Bank Digital Currencies (CBDCs)	Various nations are investigating using BC technology to develop Central Bank Digital Currencies (CBDCs), aiming to modernize their financial systems and improve overall efficiency.
2021	Non-Fungible Tokens (NFTs)	The emergence of Non-Fungible Tokens (NFTs) on BC platforms enables tokenization and ownership of unique digital assets such as artwork and collectables.
2022	Decentralized Finance (DeFi)	The rapid growth of decentralized finance (DeFi) applications on BC platforms, offering various financial services such as lending, borrowing, and trading without intermediaries.

nology with key developments and corresponding years [86], and the evolution of the same is shown in Fig. 6. BC technology enhances security, transparency, and efficiency in managing IoT devices. It utilizes BC's decentralized framework, secure record-keeping, and cryptographic measures to ensure reliable interactions, automated operations, and precise data handling among IoT devices. As systematic research findings demonstrate, BC's properties are notably advantageous in IoT device networks. As discussed in Section 1.1, IoT systems face numerous challenges, including the heterogeneity of IoT systems, insufficient interoperability, resource constraints in IoT devices, and issues related to security and privacy vulnerabilities. BC technologies offer a valuable solution to these challenges by enhancing interoperability, strengthening privacy measures, and bolstering security in IoT systems. Furthermore, BC can improve the reliability and scalability of IoT systems [87]. Table 4 demonstrates the primary traits of BC and its potential impact on IoT device networks.

Improved interoperability for IoT networks can be achieved by integrating BC. This enhancement involves the transformation and storage of IoT data into BCs. Various types of heterogeneous IoT data undergo conversion, processing, extraction, and compression and are ultimately stored in BCs. Furthermore, the improved interoperability extends to seamless navigation through diverse, fragmented networks, as BCs are built on a P2P overlay network that facilitates universal internet access.

Enhanced security for IoT systems can be achieved through BC mechanisms. One aspect involves securing IoT data by storing it as BC transactions, which undergo encryption and receive digital signatures from cryptographic keys, such as the elliptic curve digital signature algorithm [43]. Additionally, integrating IoT systems with BC, such as smart contracts, improves IoT system security. This integration facilitates automatically updating IoT device firmware to address vulnerabilities and enhance overall system security [86].

The traceability and dependability of IoT data can be significantly enhanced through BC. Data stored in BCs can be identified and verified from any location and at any time. Additionally, all historical transactions recorded in BCs are easily traceable. For instance, a BC-based product traceability system developed in [88] offers traceable services to suppliers and retailers. This system enables the inspection and verification of product quality and originality. Furthermore, the immutability of BC ensures the effectiveness of IoT data, as it becomes incredibly challenging to alter or falsify any transactions stored within the BC. BC technologies can enable self-governing interactions among IoT systems. This implies that IoT devices or subsystems can engage in automatic interactions. For instance, the study presented in [89] introduces the concept of Distributed Autonomous Corporations (DACs) to automate transactions, eliminating the need for traditional roles such as governments or companies in the payment process. By utilizing smart contracts, Decentralized Autonomous Corporations (DACs) can function autonomously, eliminating the requirement for human involvement, thereby reducing costs.

Increased reliability and scalability are additional benefits of combining BC with IoT. BC's decentralized nature removes single points of failure, significantly enhancing the reliability of IoT systems. As each node in the network has a copy of the entire BC, the system can continue to operate effectively even if some nodes fail. Moreover, BC can handle large volumes of transactions simultaneously, thus supporting the scalability needed for extensive IoT networks.

Cost efficiency can be achieved through the automation and transparency that BC provides. Operational costs can be significantly lowered by reducing the need for intermediaries

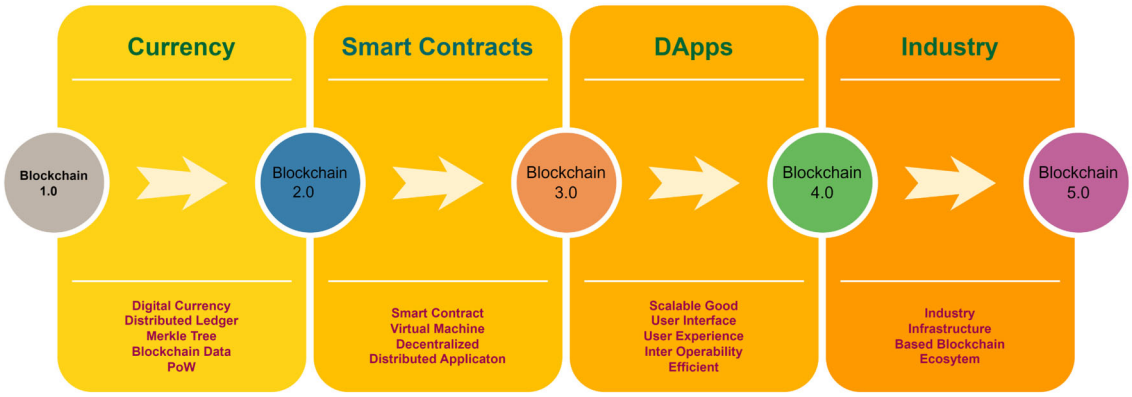


Fig. 6 Progression of BC Innovation

Table 4 Primary Traits of BC and Potential Impact on IoT Device Networks

Characteristics of BC	Supplement on IoT Device network
Decentralization	Eliminates single points of failure, enhances autonomy.
Immutability	Ensures device data integrity and tamper-proof record.
Security	Provides cryptographic identity, access control, and secure transactions.
Transparency	Enables real-time monitoring and trustworthy data sharing.
Smart Contracts	Automates device actions, updates, and access control.
Data Integrity	Safeguards data accuracy and reliability.
Peer-to-Peer	Allows direct device-to-device interactions and transactions.
Efficiency	Optimizes processes reduce intermediaries and automate tasks.

and enabling automated processes through smart contracts. This mainly benefits IoT ecosystems where numerous transactions and interactions occur regularly. BC can save costs through automated billing, supply chain management, and maintenance schedules.

Enhanced data integrity is another critical benefit. Since BC ensures that data entries are immutable and cryptographically secure, the integrity of data collected from IoT devices is guaranteed. This is crucial for applications requiring high data accuracy and trust, such as healthcare, finance, and critical infrastructure monitoring.

Privacy protection is improved as BC allows for secure, anonymous transactions. This is particularly important for IoT applications dealing with sensitive information. BC's use of public and private keys ensures that data can be shared securely without compromising user privacy. Additionally, BC can support compliance with regulations like GDPR by providing transparent and immutable records of data handling practices.

Decentralized decision-making can be facilitated through BC. In IoT networks, decisions can be made at the edge, reducing latency and improving response times. For instance, BC can enable smart contracts that allow devices to negotiate and make decisions autonomously based on pre-defined criteria, which benefits applications like autonomous vehicles and smart grids.

Improved efficiency in data management and operations is another benefit. BC can streamline data collection, storage, and retrieval processes, ensuring IoT devices operate efficiently. This efficiency extends to reducing redundancy, as BC provides a single source of truth, eliminating the need for multiple copies of the same data. In conclusion, integrating BC with IoT addresses many challenges IoT systems face and provides a robust framework for future developments. Combining these technologies can lead to more secure, reliable, and efficient IoT ecosystems supporting various applications and services.

3 Utilizing BC in IoT device network

This section presents a solution framework for implementing BC technology in IoT device networks, a growing interest and importance topic. Within this context, we advocate for a decentralized approach, where IoT devices interact directly with the BC network, eliminating the reliance on centralized intermediaries. Given the complexities of IoT network architectures, the imperative need for seamless data exchange among nodes, the inherent challenges and risks associated with data transmission, as well as the trust deficits among nodes, BC emerges as a powerful tool to address these issues and enhance the network and security of IoT devices [90]. BC technology harnesses authentication, traceability, and transaction security within an inherently untrusted environment. It facilitates real-time data handling, enabling the monitoring and regulation of data in a virtual realm. The adoption of BC technology results in heightened efficiency with faster response times, increased rates of secure data transmission, and the removal of constraints. Collectively, these advantages significantly enhance the network and security of IoT devices. Moreover, BC technology instils a decentralized trust model, allowing direct interactions and communications with IoT network devices to establish and preserve trust and data integrity. It ensures that records are immutable, making data on the BC network tamper-resistant. This, in turn, fortifies trust and bolsters the security of IoT network devices. We implement cryptographic identities for each device to fortify the security of IoT devices. Each IoT device possesses a unique crypto-

graphic identity on the BC, rendering unauthorized devices exceptionally challenging to infiltrate the IoT network or tamper with IoT data. Furthermore, to safeguard sensitive IoT data, we employ data encryption techniques, ensuring that data is securely stored on the BC and impervious to unauthorized access [91]. BC technology empowers device owners with greater control over their data, fostering user-centric data control. BC facilitates the granting and revoking of permissions for data access and sharing without reliance on third-party intermediaries. Additionally, the direct interaction between devices and the BC streamlines operations, eliminating latency associated with intermediaries. Smart contracts on the BC automate various facets of the IoT device network, such as firmware updates, access control, and data sharing, reducing the need for manual intervention and intermediaries. Blockchain technology ensures a high level of transparency and auditing in IoT device networks. It records device interactions and transactions in real-time, providing a transparent audit trail. This not only enhances transparency but also simplifies compliance with regulatory requirements and reporting, instilling a sense of accountability and integrity in the system. BC technology ensures IoT network traceability by tracking the provenance of IoT devices throughout the supply chain, enhancing authenticity and traceability. Its decentralized nature enhances resilience by eliminating single points of failure, ensuring continued operation even in adverse conditions. BC technology offers a robust and adaptable solution for IoT device networks, providing operational advantages. Various blockchain networks and platforms enable seamless communication among IoT devices, facilitate interoperability, and eliminate vendor lock-in. By removing intermediaries, blockchain technology reduces costs associated with device networks, data handling, and access control. It also addresses energy efficiency, making it suitable for energy-constrained IoT devices. These operational advantages instil confidence in the system's ability to meet diverse needs and challenges, offering potential cost savings and efficiency improvements. By [92], the application of BC can build digital solutions that enhance the real-time exchange of operational data across networks, offering a competitive advantage. Security, as highlighted by [93], is another significant benefit of BC technology, as it provides a shared, immutable ledger with hardcoded rules, reducing or eliminating the need for internal audits and third-party involvement. BC technology allows users to choose between anonymity and identity verification, ensuring secure data sharing and fostering trust. All participants can view information on the BC, and it cannot be unilaterally altered, thereby building trust and reducing the potential for fraud. This decentralized model, wherein IoT devices interact directly with the BC network, significantly enhances security, transparency, and efficiency while minimizing reliance on centralized control points and intermediaries. This approach aligns seamlessly with trustless and peer-to-peer interaction principles, paving the way for a promising future in IoT device networks. (Figure 7) shows the structure of utilizing BC in IoT device networks.

3.1 What novel approaches can be developed to ensure efficient and resource-friendly data synchronization and validation in IoT device networks? (RQ1)

Innovating novel approaches for efficient data synchronization and validation with the Blockchain is of utmost importance in low-power and resource-constrained IoT networks. One promising avenue is delegating some validation processes to edge devices, allowing local verification before data is committed to the Blockchain. By enabling these devices to perform lightweight validations and consensus, unnecessary overhead on the main blockchain network can be reduced. Moreover, selective data logging strategies can be employed to conserve valuable resources. Rather than recording every piece of data, IoT devices could prioritize and log only critical information, optimizing data transmission and storage. Batch processing techniques can also be embraced, where multiple data transactions are grouped together and submitted to the Blockchain in a single energy-efficient operation, reducing the frequency of power-intensive transactions. Sharding, which involves partitioning the blockchain network into smaller, more manageable segments, can significantly enhance resource utilization in IoT environments. Zhu and Badr [94] This approach allows subsets of IoT devices to work on specific shards of the Blockchain, reducing the computational burden on individual devices. Furthermore, designing lightweight consensus algorithms tailored to the unique constraints of IoT networks can further optimize resource usage. These algorithms can be energy-efficient and accommodate IoT devices' limited processing power and memory capacity. Data compression methods can play a pivotal role in enhancing efficiency. Implementing efficient compression techniques before data is written to the Blockchain can reduce the amount of data transmitted and stored. Wadhwa et al. [95] This conserves energy and minimizes the strain on storage resources. Hierarchical blockchain structures and sidechains offer promising solutions for segregating data based on priority and importance. IoT devices can choose where to store and validate information by categorizing data into different tiers. Less critical data can be stored on sidechains connected to the main Blockchain but have reduced resource requirements. This approach ensures that vital data undergoes rigorous validation on the

main Blockchain, while less critical data can be handled with fewer resources on sidechains. Another innovative avenue involves using threshold signatures for collaborative validation among IoT devices. Instead of each device independently validating transactions, a group of devices collaborates to verify transactions collectively. This reduces the energy overhead for individual devices and leverages the collective processing power of the network. Machine learning-based validation models can adapt to changing network conditions, further optimizing validation processes. These models can intelligently adjust validation criteria based on resource availability and network congestion, ensuring efficient and timely validation. Exploring energy harvesting mechanisms, where IoT devices generate power from their surroundings, can alleviate some power limitations. Solar panels, kinetic energy harvesters, and other renewable energy sources can be integrated into IoT devices, allowing them to operate for extended periods without relying solely on battery power. Additionally, implementing resource trading mechanisms within IoT networks can enable devices to share resources like processing power and bandwidth. Devices with excess resources can provide computational support to resource-constrained devices in exchange for tokens or incentives, fostering collaboration and resource optimization. Scheduled blockchain access for IoT devices can also help manage power constraints. Devices can be programmed to access the blockchain network during periods of lower power consumption, ensuring that critical transactions are processed while conserving energy. Lastly, Blockchain as a Service (BaaS) platforms tailored to IoT environments can provide off-the-shelf solutions that align with resource constraints [96]. These platforms offer preconfigured blockchain setups optimized for IoT, simplifying the integration process and ensuring that IoT and blockchain technologies work seamlessly within the resource limitations of the network. In summary, a combination of local validation, selective data logging, batch processing, sharding, lightweight consensus algorithms, data compression, hierarchical structures, collaborative validation, machine learning, energy harvesting, resource trading, scheduled access, and BaaS platforms can collectively pave the way for efficient and resource-friendly data synchronization and validation between IoT devices and the Blockchain in low-power IoT networks. Table 5 shows the Comparative analysis of resources for data synchronization and validation between IoT devices and Blockchain.

3.2 Is BC a viable solution for addressing IoT's technical and security challenges?(RQ2)

Yes, BC provides a secure, tamper-resistant way to store and manage data. Its decentralized and cryptographic nature makes it difficult for malicious actors to compromise the integrity of IoT data. The following BC-based solutions which are surveyed, defined below one by one:

3.2.1 BC IoT security issues

According to Sezer [103], the deployment of IoT devices is hindered primarily by security concerns. Prominent challenges center on attacks that impede users' access to network resources, disrupt system communication, engage in espionage, compromise personal data, and deploy botnets for cryptocurrency mining [104]. A notable difficulty emerges when numerous computer systems overwhelm a centralized server with an excessive volume of data requests, leading to a denial of service for the intended clients of the system. Insecure IoT devices introduce vulnerabilities that cybercriminals exploit, enabling them to infiltrate systems and initiate distributed denial-of-service attacks. BC provides a more robust level of encryption than IoT, preventing parties from overwriting existing records on the network [105]. Integrating IoT data into the BC introduces an additional layer of security, preventing cybercriminals from easily accessing the network [106]. Viriyasitavat et al. [11] highlight that BC's encryption and distributed storage features provide a secure means of recording data in IoT devices, enabling detailed transactions without human intervention. This process ensures data integrity, building trust among all stakeholders in the supply chain. Each participant in BC technology receives a unique identity linked to their account, allowing only the authorized owner to execute transactions. The robust encryption within the BC makes it challenging for hackers to disrupt or compromise the established chain [11]. The oversight of all transactions by miners further maintains the integrity of the BC system. Once a block or transaction is added to the BC, it becomes immutable for security purposes. The track record of hackers failing to breach or compromise BC underscores its trustworthiness, tamper-proof nature, and resilience against technical failures and malicious attacks [107]. This resilience is achieved through the decentralized nature of BC.

3.2.2 Challenges related to centralization in BC and IoT

The predominant issue in IoT systems lies in their centralized client-server structure, making them susceptible to failures. BC addresses this concern by distributing requests across all devices within the network. Despite advancements, centralization remains a challenge for IoT, mainly due to sensor networks' substantial volume of data, potentially burdening systems with low processing speeds [108]. Authentication, authorization, and the network of numerous users connecting

Table 5 Comparative analysis of resources for data synchronization and validation between IoT devices and Blockchain

Author	Year	Consensus Approach	Energy Consumption	Memory Utilization	CPU Utilization
H. Gao et al. [97]	2018	Reinforcement learning approach for MEC system	Increasing energy consumption	–	Increasing computation capacity
Y. Liu et al. [98]	2020	DRL approach	Energy consumption increases with the number of cells	–	CPU utilization increases with the number of cells
Wei Hu et al. [99]	2021	P2P DE transaction model	Less time for energy exchanges	–	–
Wadhwa S et al. [96]	2022	PoW mechanism	21%	24%	–
M. A. A. Ghamdi et al. [100]	2022	SDN-BC classifier	50% less energy between BCF, ASCIR, and BSDNSR	Depends on file size	CPU utilization increases with attack rate
A. Sasikumar et al. [101]	2023	Proof-of-importance mechanism	Reduces with increasing devices	–	Increases with increasing devices
A. Salh et al. [102]	2023	Alternative Direction Algorithm at every iteration of FL time	Maximum power consumption	–	No change

to the centralized server create congestion issues. To overcome this, Alam [109] recommends substantial investments in robust centralized servers capable of accommodating the extensive flow of data within the network. In the context of IoT networks handling transactions across diverse devices from various organizations, tracing the source of information leaks in the event of an attack becomes challenging. Additionally, identifying the owners of the generated data could be improved. As per Popescul and Georgescu [110], in IoT technology, unauthorized access to the server is sufficient for a hacker to manipulate or delete data. Conversely, in BC, data is shared across multiple nodes. A distributed system ensures that valid modifications to information are disseminated among various network participants. In scenarios involving substantial requests, an IoT network server may experience breakdowns. In contrast, by distributing requests across the entire network instead of a specific node, a distributed ledger exhibits significant resistance to network pressure. The failure of an IoT network server leads to a network-wide failure, inconveniencing users. Nevertheless, a BC network with many interconnected devices is only susceptible to such failures if the capacity of the devices is at most that of the network.

Ghuli et al. [111] identified challenges in connecting to a remote server in IoT networks, a concern BC addresses by allowing clients to choose paths for information dissemination. The IoT server bears responsibilities such as storing, receiving requests, and managing data, which may lead to potential overload during peak periods. In contrast, BC distributes its load across multiple participants, facilitating trusted data sharing without intermediaries and removing third-party involvement in value chain transactions. As Mohanta et al. [112] highlighted, BC's Smart contracts enable automated payments and transactions among diverse IoT devices. By preserving immutable historical records of IoT devices without centralization, BC enhances the performance of these devices on the network. Enterprises can employ BC for the storage of smart contracts, facilitating the execution of contractual agreements among involved parties based on predefined criteria. For instance, contract payments can be automatically authorized without requiring third-party intervention [113]. In BC technology, transaction data is distributed across a network of devices, eliminating the presence of a singular managing node. This absence of a central vulnerability point enhances the resilience of data against hacking, theft, and forgery. The failure of an individual node does not impede the functionality of other nodes within the network, fostering trust in BCs during transaction processes. Upon reaching a consensus on a transaction, the system broadcasts the information to multiple network nodes, diverging from the reliance on a singular server, as seen in traditional transaction systems [114]. Given the openly distributed nature of transactions, this methodology is a preemptive measure against cyber-attacks, theft, and other malicious activities. The decentralized architecture of BC obviates the necessity for third-party involvement through robust verification and validation mechanisms. Every transaction recorded in the distributed ledger of the BC is accessible to all participants in the network, ensuring transparency and fostering trust throughout the system. Consequently, BC technology vigilantly monitors all transaction data, rendering it highly resistant to fraudulent activities and challenging to decipher.

3.2.3 Challenges related to transparency in BC and IoT

BC's inherent transparency is a foundational characteristic that allows authorized entities to access the network and scrutinize past transactions. This capability facilitates the swift identification of data leakages, enabling immediate corrective measures before potential exploitation by hackers [115]. The transparent record-keeping nature of BC empowers authorized entities to monitor and analyze ongoing network activities. In organizations dealing with extensive data transactions spanning multiple networks, the unalterable ledger function acts as a protective measure, tracing the path of data or goods throughout the nodes in the supply chain [116]. BC provides heightened security for data originating from IoT sensors, elevating its status as a valuable ledger. The nodes within BC technology exhibit similarities to objects in IoT systems, reinforcing the security and reliability of interconnected objects. IoT systems based on BC streamline business transactions, fostering transparency for an enhanced client experience. The study by Ibarra-Esquer et al. [117] affirms that integrating BCs into IoT networks facilitates secure and distributed messaging among devices, with BC treating messages akin to handling financial transactions on the Bitcoin network.

The capability of BC to offer networks based on smart contracts, exemplified by platforms like Ethereum, provides IoT devices with the confidence to interconnect and engage in trade, ensuring that transactions will adhere to predetermined rules. The distributed ledger feature of the BC system guarantees the integrity of transactions, eliminating the possibility of interference and addressing trust concerns among stakeholders [116]. Notably, BC demonstrates faster processing of large transaction volumes than IoT systems and exhibits an efficient network of millions of connected devices, even as their numbers increase. The inherent trustworthiness of BC allows organizations to reduce processing overhead costs

associated with IoT gateways, including communication and hardware overheads.

3.2.4 Challenges related to interoperability in BC and IoT

Interoperability arises when two or more disparate systems or devices within networks communicate to achieve a common objective. In the context of IoT, systems and devices often need more communication protocols, standardized data formats, and cohesive technologies, resulting in fragmentation. This fragmentation hinders the seamless sharing of data among interconnected devices. Many IoT devices are specifically designed to function within predetermined hardware configurations, creating obstacles for the smooth integration of new devices and potentially giving rise to operational challenges. As highlighted by Amjad et al. [118], interoperability challenges in IoT devices become evident when they communicate services to the cloud or attempt to consolidate data from different devices. Moreover, interoperability issues can arise when other devices try to reuse services generated by another IoT device. BC, leveraging cross-chain technology, possesses the capacity to exchange information across various systems, devices, and networks. Cross-chain technology is designed to achieve interoperability, enabling seamless communication between private networks or between public BC and private networks. Additionally, BC incorporates tools such as atomic swaps and multi-chain protocols to enhance and streamline interoperability processes.

3.2.5 Resilient attributes of BC

The scalability of an IoT device pertains to its ability to adapt to environmental changes and meet future demands [42]. In the context of BC, scalability refers to the system's capability to handle a higher volume of transactions per second compared to other systems, a critical characteristic for managing increasing workloads [46]. In the BC domain, scalability involves supporting many transactions and nodes within the network. Scalability is of utmost importance in IoT since the BC system is distributed, and the verification of transactions in each node holds significant importance. The regulation of the number of transactions in each node is essential. As Reyna et al. [119] highlighted, the transaction speed may become impractical for users, and introducing more active participants could further slow down transaction speeds. BC scalability is affected by network congestion, similar to centralized systems where transaction completion depends on network load. To address BC scalability challenges, researchers propose conducting transactions outside the BC and storing and accessing information within it. Additionally, approved networks can be leveraged to tackle scalability issues in BC technology. Effectively managing scalability is crucial for the success of IoT, and it can be achieved by implementing efficient, robust, and standardized services on the device.

3.2.6 Scalability in the Context of BC and IoT

An IoT device is considered scalable when it can adapt to environmental changes and meet future requirements effectively [42]. In the context of BC, scalability refers to the system's ability to handle more transactions per second than other systems, making it a crucial characteristic for systems capable of managing growing workloads. In the context of BC, scalability denotes the platform's ability to accommodate an increasing volume of transactions and an expanding number of nodes within the network. When applied to IoT scalability within a BC system, the distribution and verification of transactions across each node become pivotal. It is crucial to regulate the number of transactions in each node carefully. According to Reyna et al. [119], the practicality of increasing the number of active participants is hindered by transaction speed limitations, with procedures taking several hours for a user. Introducing more participants into the BC network tends to decelerate transaction speeds, presenting scalability challenges reminiscent of centralized systems where transaction completion is contingent on network congestion. To address the scalability challenge in BC, researchers propose conducting transactions outside BC and storing and accessing information within it. Additionally, approved networks can be employed to resolve scalability issues in BC technology. Scalability is a pivotal factor for the success of IoT, and it can be effectively managed through efficient, robust, and standardized services on the device.

3.3 Impact areas of BC utilization in IoT device network?(RQ3)

According to the literature review, Blockchain is applied in four primary domains within the network of IoT devices. As IoT devices connect, a vast amount of data is generated and centralized in traditional IoT systems. To secure this data, we have to secure IoT devices by using BC technology. As earlier, we have discussed the significant challenges in IoT networks. We have reviewed several research papers and found various impact areas where BC technology uti-

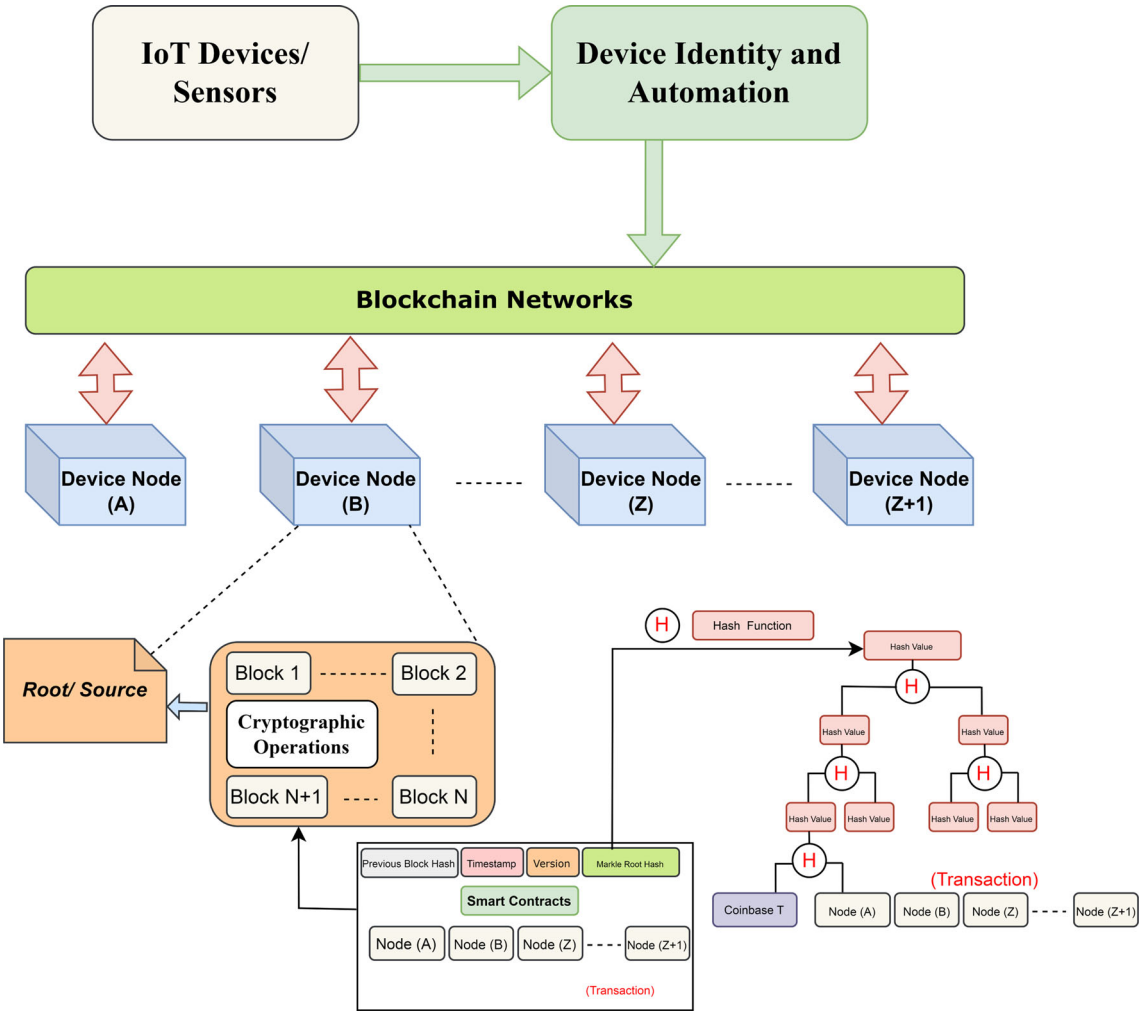


Fig. 7 Structure of Utilizing BC in IoT device networks

lization is most important in IoT networks [120]. Figure 8 represents the most impactful areas of BC technology.

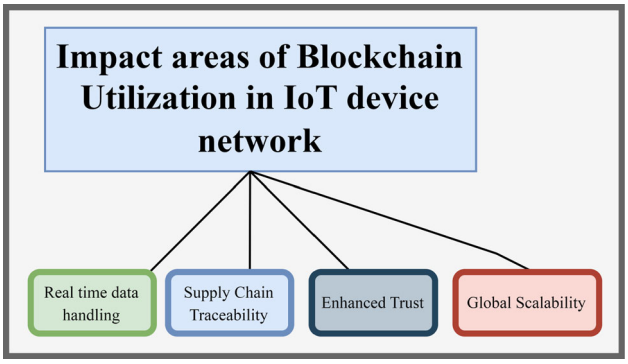


Fig. 8 Represents the impact areas of BC Utilization in IoT device networks

3.3.1 Real-time data handling

When we monitor real-time data of IoT devices, data handling is challenging. A real-time network of data from IoT devices using blockchain technology involves the seamless processing, monitoring, and regulation of data generated by these devices, all promptly. This approach is instrumental in ensuring that data is handled as effectively as it is produced, thereby enhancing the responsiveness and effectiveness of IoT ecosystems [120]. IoT devices are at the heart of this system, each equipped with sensors and data collection capabilities. These devices continuously gather a wide range of data, including variables such as temperature, humidity, location, and other relevant metrics. The data collected by IoT devices undergoes a secure transmission process, typically through gateways or edge devices responsible for initial data preprocessing. These edge devices are

critical in reducing latency and filtering non-critical information before transmitting the data to the BC network. The BC network comprises a distributed set of nodes, each tasked with processing and validating real-time data transactions. The collaboration of these nodes ensures the maintenance of a tamper-resistant ledger, guaranteeing the security and integrity of data. Smart contracts deployed on the BC introduce automation depending on predefined conditions. These contracts execute real-time actions when specific data triggers predetermined rules [121]. For instance, a smart contract may activate alerts or initiate responses when an IoT sensor detects critical thresholds or events. Users interact with the system through user interfaces, dashboards, or monitoring systems that provide real-time visibility into the IoT ecosystem. This real-time monitoring empowers users to monitor device statuses and data streams closely. The network's security is paramount and ensured through robust measures. These measures encompass data encryption, access control mechanisms, and authentication processes, all working to safeguard data privacy and maintain network integrity. Blockchain networks' scalability is essential to accommodate the growing volume of IoT data efficiently. To this end, energy-efficient consensus mechanisms like Proof of Stake (PoS) are employed, making it possible for IoT devices with limited power resources to participate actively in the network. For IoT applications involving supply chains, blockchain integration provides end-to-end traceability and provenance tracking for IoT devices, enhancing authenticity and traceability [122]. Advanced analytics and machine learning models are applied to the data in real time to derive actionable insights and predictions, further strengthening the system's utility. Finally, the blockchain ledger simplifies compliance with regulatory requirements by furnishing a transparent audit trail of all transactions and data interactions, ensuring transparency and accountability. By implementing this real-time data handling system using blockchain, IoT devices can operate efficiently and securely while responding to data promptly. This approach is adaptable to various IoT applications, including smart cities, industrial automation, healthcare, and more.

3.3.2 Supply chain traceability

Supply chain traceability using blockchain in IoT device networks is leveraging blockchain technology to create a transparent and tamper-proof record of an IoT device's journey through the supply chain [116]. Each IoT device is assigned a unique cryptographic identity on the blockchain, and every step in the device's lifecycle, from manufacturing to deployment, is securely and immutably recorded. This traceability ensures stakeholders can easily verify the device's authenticity, origin, and compliance with quality standards. Blockchain's real-time updates and secure documentation streamline auditing, reduce the risk of counterfeit products, enhance trust, and ultimately contribute to a more efficient and secure IoT device network within the supply chain. Table 6 demonstrates the blockchain technology's supply chain traceability of IoT device networks [123]. This table outlines various stages and activities within the supply chain for IoT device networks, showcasing how blockchain technology contributes to traceability, transparency, and trust-building at each step.

Table 6 Various Stages and Activities within the Supply Chain for IoT Device Network

Stage	Activity	Description
Manufacturing	Device Production	Unique cryptographic identity assigned.
	Quality Testing	Test results recorded on the BC.
Distribution	Shipping	Shipment details and location recorded.
	Ownership Transfer	Change of ownership securely documented.
Retail Deployment	Inventory network	Real-time tracking of device availability.
	Device Activation	Activation details recorded on BC.
	End-User Interaction	User interactions with devices recorded.
Compliance	Recalls	
Trust Building	Regulatory Audits	Auditing simplified with BC data.
	Product recalls	Swift recall of affected devices enabled.
	Origin Verification	Authenticity easily verified by stakeholders.
	Anti-Counterfeit Measures	Reduced risk of counterfeit products.
Efficiency	Reduced Intermediaries	
	Supply Chain Integration	Cost reduction through BC efficiency.
		International supply chain transparency.

3.3.3 Enhanced trust

Enhancing trust in IoT devices through the application of blockchain technology represents a significant leap forward in addressing the intricate challenges associated with managing and securing these interconnected devices. At the core of this transformation lies the concept of a unique and immutable digital identity for each IoT device. Blockchain technology enables the secure allocation and storage of these cryptographic identities, creating an indelible digital fingerprint for each device. This fingerprint is both resistant to tampering and verifiable, ensuring that only authenticated devices are granted access to the network. Moreover, it establishes a robust foundation for tracking and verifying device ownership, origin, and compliance with stringent quality standards [143]. The linchpin in bolstering trust within IoT devices is the establishment of a transparent and incorruptible ledger on the blockchain. Every transaction, interaction, and IoT device data exchange is meticulously recorded in real time within this ledger. Each entry is immutable and invulnerable to alteration, providing all authorized participants within the ecosystem with an unobstructed view of device activities and data flows. This transparency engenders unwavering confidence among stakeholders, as they can rely upon the accuracy and trustworthiness of the information at their disposal. Furthermore, blockchain technology introduces the groundbreaking concept of smart contracts, autonomous agents automating various facets of IoT device networks. These self-executing contracts emerge based on predefined conditions, such as specific data thresholds or access requests. By minimizing the necessity for manual intervention, smart contracts diminish the potential for human errors and elevate the efficiency and reliability of device network processes. Decentralization represents another pivotal dimension of BC's contribution to enhanced trust. BC operates as a decentralized network, obviating the need for central authorities or intermediaries. Decentralization confers a heightened level of resilience upon the ecosystem by eradicating single points of failure and reducing the susceptibility to manipulation or interference by a central authority. Mohanta et al. [144]Trust is no longer vested in a singular entity but is instead founded upon the collective integrity of the entire network. Supply chain traceability is another facet of BC's profound impact on trust within IoT devices. By furnishing an exhaustive record of each device's odyssey through the intricate supply chain, blockchain provides unequivocal assurance regarding the authenticity of devices. Simultaneously, it curtails the potential for counterfeit or unauthorized products to infiltrate the ecosystem. BC's capabilities further solidify trust among stakeholders in scenarios necessitating swift and efficient recalls due to safety concerns. In addition to these core elements, blockchain technology integrates a formidable array of security measures, including robust encryption, multi-factor authentication, and vigilant intrusion detection systems, assiduously safeguarding data and transactions. Ali et al. [145] The empowerment of device owners through user-centric data control offers the ability to grant or revoke permissions for data access and sharing without reliance on third-party intermediaries. Compliance and auditing obligations are markedly streamlined through BC's inherently transparent and auditable ledger, ensuring that regulatory requirements are met and fostering a climate of accountability. The promotion of interoperability across diverse IoT devices and platforms facilitates seamless communication, instigating a greater degree of flexibility and mitigating the risks of vendor lock-in. In summation, the multifaceted approach of blockchain technology to enhance trust within IoT devices [146] heralds a transformative era in device networks and security. It ushers in an era characterized by transparency, data integrity, robust security, and streamlined automation within the IoT ecosystem. The foundation of trust established by blockchain technology not only enhances the security of IoT devices but also opens the door for their secure and efficient operation across diverse industries and applications. This spans from smart cities and industrial automation to healthcare and beyond. Figure 9 shows the enhanced IoT device network.

3.3.4 Global scalability

Global scalability in IoT device networks is critical in today's era of rapid digital transformation and the proliferation of interconnected devices worldwide. The integration of BC emerges as a groundbreaking solution to tackle the growing challenges of managing IoT devices globally. At its core, Blockchain's decentralized architecture is inherently designed to facilitate global scalability [147]. As the number of IoT devices continues to soar globally, the blockchain network can dynamically adapt by incorporating additional nodes, ensuring that the network remains responsive and efficient, regardless of the geographic distribution of these devices. This adaptability is crucial because IoT deployments often span remote and diverse locations, ranging from smart cities and industrial complexes to agricultural fields and healthcare facilities [3]. A significant contributing factor to BC's global scalability is its energy-efficient consensus mechanisms, prominently exemplified by PoS. Considering numerous IoT devices operate with constrained power resources like batteries or energy-harvesting solutions, PoS is notably more energy-efficient than conventional alternatives like PoW [94]. This energy-conscious approach enables IoT devices to actively engage in the blockchain network without quickly exhausting their power reserves. Such energy efficiency is indispensable for promoting the global network of IoT devices, even in regions with challenging or off-grid power infrastructures. Blockchain's emphasis on interoper-

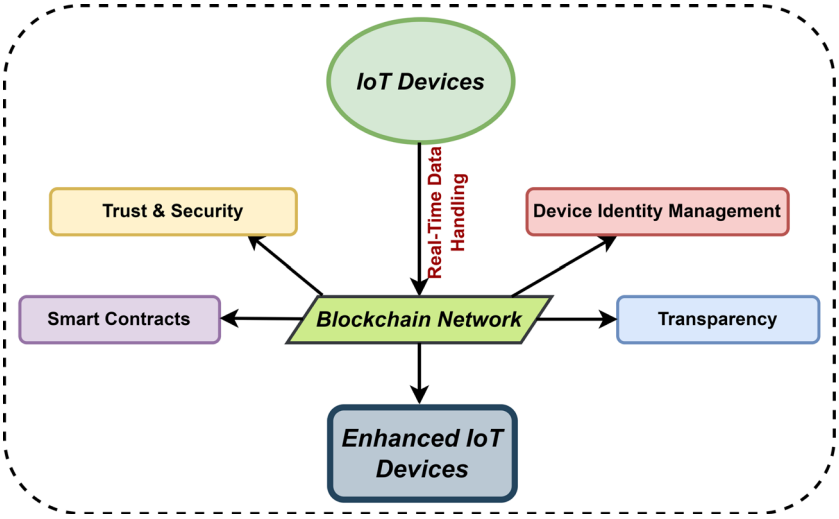


Fig. 9 Enhanced IoT device network

ability plays a pivotal role in achieving global scalability. IoT ecosystems encompass a multitude of devices from various manufacturers and ecosystems. Here, blockchain networks serve as the unifying platform, allowing IoT devices to seamlessly communicate and operate within the same ecosystem, regardless of their origins. This eliminates vendor lock-in and fosters a harmonious and integrated global IoT landscape. Cost efficiency constitutes another compelling factor contributing to global scalability. Blockchain's ability to eliminate intermediaries and automate tasks through Smart contracts significantly reduces the expenses related to provisioning, data handling, access control, and data sharing. This cost-effectiveness makes it economically feasible to manage many IoT devices distributed across the globe, facilitating scalability without incurring excessive overheads. In the context of a global IoT device network, security and trust are of utmost importance. With the diverse stakeholders and the international flow of devices and data, blockchain's inherent security features, including a cryptographic identity network and tamper-resistant ledgers, establish a robust foundation for trust. This trust extends across international boundaries as devices and data interact with various entities, encompassing manufacturers, users, and regulatory bodies. The emphasis on security and trust reassures the audience about the reliability of blockchain technology and its ability to protect sensitive data. Blockchain technology offers a multifaceted and comprehensive solution for achieving global scalability in IoT device networks. Its decentralized architecture, energy efficiency, interoperability, cost-effectiveness, and robust security features position it as a catalyst for efficiently and securely managing IoT devices across diverse regions, industries, and applications worldwide. As the number of IoT devices continues to surge globally, blockchain's adaptability and resilience are indispensable for ensuring this ever-expanding ecosystem's seamless, reliable, and sustainable network.

3.4 Identify the weaknesses and attack vectors unique to BC-integrated IoT networks.(RQ4)

BC integrated IoT device network systems offer enhanced security, transparency, and automation but are not immune to potential weaknesses and attack vectors that threaten their integrity. It is crucial to identify these vulnerabilities and explore strategies for countering them through advanced security measures and comprehensive threat modeling. One significant weakness in Blockchain-integrated IoT systems is the potential for 51% attacks. Blockchain networks are secured through consensus mechanisms, and in PoW systems, an attacker with over 50% of the network's computational power can manipulate transactions. In the context of IoT, if a malicious entity gains control over the majority of nodes, it can compromise the integrity of device network processes. To counter this, implementing more robust consensus mechanisms such as PoS or Delegated DPoS can reduce the risk of 51% attacks [81], as they rely on ownership and stake rather than computational power. Table 7 defines the attack vectors on the IoT network. Vulnerability is the exposure of private keys. IoT devices often have limited security capabilities, making them susceptible to physical attacks or exploits. If an attacker gains access to the private keys used for authentication and transactions, they can impersonate or manipulate data. Using hardware-based security, such as Trusted Platform Modules (TPMs) or Hardware Security Modules (HSMs), can safeguard private keys and protect against physical attacks. Smart contract vulnerabilities are also a concern. Flaws or vulnerabilities in smart contracts can lead to unauthorized

Table 7 Attacks on IoT Devices

Company Name	Category	Year	Description
Man-In-the-Attack [124]	External	2019	It is done on the internet between two point nodes to intercept their communication, get their essential data, and monitor it.
Denial of sleep attack [125]	Internal	2019	This attack aims to exhaust the batteries of the devices by increasing their cycle or by forcing nodes to wake at an unnecessary time by including additional duties like listening, retransmission, etc., the lifetime of the IoT network detecting unwanted usage of energy consuming tasks like idle, retransmission, overhearing, etc. and remains a challenging task.
Traffic analysis attack [126]	External	2019	Hackers do this attack. Hackers try to simultaneously enter the same network traffic and access something about you and your activity.
Physical-based attack [127]	Internal/External	2020	Attacker attacks on the physical IoT device components like Sensors, Radio-frequency identification Readers, and Radio-frequency TAG types occur because they all are powerless for physical assaults. In this attack, attackers identify all physical relations between physical equipment and IoT devices.
Routing information attack [127]	External	2020	In this attack, the attacker generates the complex network by spoofing and modifying the information of the routing protocol. The attacker aims to provide inappropriate data and information to break the network.
Sinkhole attack [128]	Internal	2021	It is an insider attack where an attacker compromises a node inside the network traffic and launches an attack. Nodes try to attract all the traffic by advertising their fake routing updates.
Side channel attack [129]	Internal/External	2021	The mitigation of this attack is to exploit the security of a channel that aims to get data and information about the program execution to influence the system by using indirect effects on the system or its hardware rather than targeting the code directly.
Object replication attacks [130]	Internal	2021	The Attacker generates a line of code and adds it to the network, a malicious line of code, to decrease the network's reliability. The task of this attack is to delete and corrupt the received data and information.
Good Mouthing Attack [131]	Internal	2021	It forces malicious nodes to have high ratings to appear trustworthy.
Ballot Staffing attack [132]	Internal	2022	Malicious nodes seek to promote another malicious node by giving favourable opinions about them to increase their chances of being faithful and trusting.
Shadow IoT [128]	Internal	2023	The rising popularity of BYOD (Bring Your Own Device) has led to an influx of personal devices, including IoT-based ones, being used in workplaces without adequate security measures. Employees often need the IT department's knowledge to bring these devices, inadvertently creating potential entry points for cybercriminals to exploit and breach the company's systems.
Supply Chain Attacks [129]	External	2023	The prevalence of supply chain attacks is rising, and IoT devices are not exempt from these security breaches. As we look ahead to 2024, there will likely be a surge in attacks targeting the vulnerabilities in the supply chain to compromise IoT devices and networks.
Lack of Network Segmentation [130]	External	2023	IoT devices commonly share networks with others, making an attack on one device possible to propagate to others. Looking forward to 2023, we should anticipate increased attacks capitalizing on the absence of network segmentation, enabling cybercriminals to infiltrate IoT devices and networks more efficiently.

access, data breaches, or financial losses. Conducting rigorous code audits, implementing formal verification methods, and using secure coding practices can mitigate these risks. [123] Additionally, regular updates and patch networks for smart contracts are essential to address vulnerabilities as they are discovered. Interoperability challenges can create attack vectors. When different blockchain networks and IoT ecosystems interact, inconsistencies or gaps in security protocols can be exploited. Developing standardized protocols and frameworks for interoperability and robust access control mechanisms can mitigate these risks. Ensuring that only authorized devices and networks can access the blockchain is essential. Moreover, IoT devices can be vulnerable to Distributed Denial of Service (DDoS) attacks, disrupting the device network and compromising the integrity of data stored on the blockchain. Implementing DDoS mitigation strate-

gies, such as traffic analysis and rate limiting, can help protect the network's availability. Thorough threat modeling is essential to counter these weaknesses and attack vectors. This involves identifying potential threats, assessing their impact, and designing security measures accordingly. A defence-in-depth strategy combining encryption, authentication, access control, and intrusion detection can fortify the security of Blockchain-integrated IoT device network systems. Regular security audits, penetration testing, and continuous monitoring are vital for identifying and addressing evolving threats. While Blockchain-integrated IoT device network systems offer numerous benefits, they are not without vulnerabilities. Identifying and countering these weaknesses through advanced security measures, robust consensus mechanisms, hardware-based security, secure coding practices, and comprehensive threat modeling is essential to ensure the integrity and reliability of these systems in the face of evolving cyber threats. Table 8 discusses the IoT-Blockchain Integration-Research addressing.

4 CASE STUDY: Smart healthcare IoT system

This section explores the case study of secure IoT device communication within healthcare. The healthcare industry has undergone a significant transformation by integrating IoT technology, facilitating seamless connectivity and access to data. This advancement involves using sensors embedded in medical devices to collect vital health data and patient information. These devices, like pacemakers, insulin pumps, and remote patient monitoring systems, play a pivotal role in managing chronic illnesses by continuously gathering essential medical information such as medical history, heart rate, diagnoses, prescriptions, and body temperature.

The connectivity offered by IoT-enabled healthcare applications allows for remote monitoring. It streamlines the coordination of physical settings, enhancing operational efficiency, improving clinical procedures, and facilitating better resource utilization to elevate the overall patient care experience. However, the interconnected nature of these devices gives rise to privacy concerns. Inadequate security measures could compromise sensitive information, making it susceptible to exploitation by malicious entities for personal gain. Additionally, handling vast amounts of real-time data might introduce inaccuracies, while security vulnerabilities within the IoT infrastructure can pave the way for intricate cyber-attacks.

4.1 Problem statement

Let's imagine a scenario in smart healthcare where different medical sensor devices measure a range of vital signs like blood pressure, oxygen saturation, heart rate, EEG, magnetic field, and temperature. Additionally, wearable devices such as fitness trackers, smartwatches, and VR headsets gather data on the patient's daily activity, heart rate, sleep patterns, and steps. These devices utilize IoT applications for monitoring and services like medication refills. Let's symbolize these sensors as $\{S_1, S_2, \ldots, S_m\} \in S$.

Each sensor collects data from the environment and transmits it to other sensors, which ensures the system's smooth functioning. Sensors communicate using the serial bus communication protocol.

So, each sensor has a unique ID from $\{I_1, I_2, I_3, \ldots, I_m\}$ allocated to each sensor for data sharing. Suppose S_i and S_j want to exchange data for task execution. Each unique sensor has its own data stream, i.e., $\{\Psi_1, \Psi_2, \ldots, \Psi_i\} \in \Psi$. A sensor S_1 has Ψ_1 data stream, S_2 has Ψ_2 data stream, and so on. A source sensor (S_i) sends the data to the receiver sensor (S_j) to help the smooth operation.

$$Si\left(\sum_{k=0}^{m}(\Psi k)\right) \rightarrow Sj, \{Si, Sj\} \in S \tag{1}$$

Suppose, an attacker ξ with a fake id ε infiltrates S_i such that,

$$\varepsilon \leftarrow S_i\{\Psi i\} \tag{2}$$

$$\varepsilon \xrightarrow{\text{Manipulates}} S_i\{\Psi i\} \tag{3}$$

$$S_i\left(\sum_{k=0}^{m}(\Psi k)\right) \xrightarrow{\text{Transmitted}} \xi \xrightarrow{\text{Manipulates}} \sum_{k=0}^{m}(\Psi' k) \tag{4}$$

$$\sum_{k=0}^{m}(\Psi' k) \xrightarrow{\text{Transmitted}} S_j \tag{5}$$

The manipulated data is then sent to the IoT network, which feeds spurious information to all sensors. These altered sensor readings could result in incorrect diagnoses or inappropriate treatments, posing serious risks to patient safety.

Therefore, there is a requirement to authenticate the ID by analyzing the network traffic behavior. Based on the above discussion, the problem is formulated as:

$$\mu : R^n \rightarrow \{I_i, \varepsilon\}. \tag{6}$$

4.2 Proposed approach

We have designed a secure method for sharing healthcare data using AI and smart contracts, leveraging the InterPlanetary File System (IPFS) benefits. In Fig. 10, patients utilize diverse wearable devices, including smart glasses, blood pressure sensors, fitness bands, etc. The wearables worn by

Table 8 IoT-BC Integration- Research addressing

Research	Application	Contribution	Privacy	Centralization	Trust	Security	Reliability	Speed
Ruminot-Ahumada N et al. [131]	Home/ factory/ healthcare	Infrastructure Model		✓		✓	✓	✓
Hafeez I, Antikainen M et al. [133]	General	Security protocols	✓		✓	✓	✓	
utun I et al. [134]	Industrial applications	BCoT architectural Model	✓	✓		✓		
Shammar EA et al. [135]	Smart Grid	Scenarios, framework					✓	
Liu H, Shen F et al. [136]	IoT ecosystem	Case study			✓		✓	
Hammi MT et al. [137]	Industrail IoT in supply chain	Design, implementation and test a device tool			✓		✓	
nezevic S. et al. [138]	Industrial IoT	Plateform design, modeling, algorithm					✓	
Boncea R et al. [139]	Time-sensitive (Healthcare)	Modeling					✓	
Panarello A et al. [140]	General	Architectural model, scheme				✓		
Mazhar N et al. [141]	Scenarios	Shared economy application		✓		✓		
uan J et al. [142]	General	Hierarchical architecture	✓	✓	✓	✓		

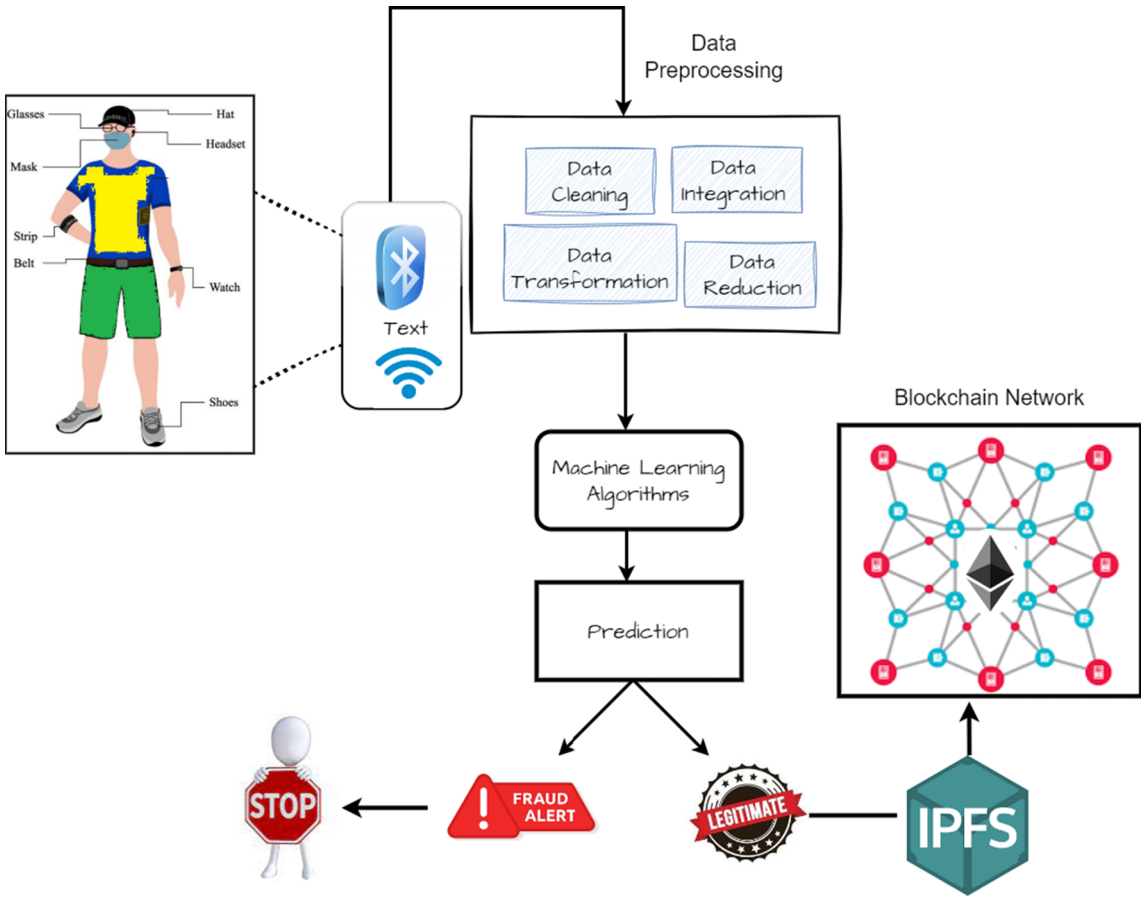


Fig. 10 Proposed System

patients establish a connection with smartphones belonging to doctors and insurance subscribers through WiFi or Bluetooth. The data produced by these wearables is transmitted to a fraud detection system. After data generation, the data preprocessing phase involves cleaning, integration, transformation, and reduction techniques. A machine learning algorithm is then employed on the preprocessed data to predict whether the data is fraudulent or legitimate. Legitimate IoT data is then stored in IPFS, and the IPFS key is stored in the BC with the smart contracts. This secure IoT data plays a crucial role in healthcare for various purposes. This secured IoT data facilitates the real-time monitoring of patient's vital signs, medication schedules, and overall health trends. Furthermore, this data is a valuable resource for medical research by offering a wealth of anonymized information for analysis.

Our approach stores data collected from various sensors in a database's comma-separated values (CSV) files. This dataset, denoted as $\mathbb{Z}$, encompasses regular and IoT attack traffic in medical IoT. The data undergoes classification in an AI layer to distinguish between attack and normal data. Data preprocessing and analysis are needed to address class imbalances before feeding it into the Machine Learning (ML) classifier. We performed pre-processing tasks utilizing Pandas, Numpy, and Python libraries. This involved normalizing the data, addressing missing values, and mitigating the class imbalance issue within the dataset. Data preprocessing involves several steps: cleansing, selecting instances, normalization, one-hot encoding, modification, feature extraction, and selection. This process aims to identify missing values, noisy or invalid data, and handle normalization concerns. Specifically, we employ standardization scaling, utilizing mean and standard deviation to normalize the data. Next, a variety of AI-based classifiers, including Random Forest (RF), Gaussian Naive Bayes, Decision Tree, Logistic Regression, and KNeighbors, are applied to categorize data into adversarial and non-adversarial categories effectively. Subsequently, the performance of these classifiers is assessed using accuracy metrics. Following this evaluation, non-adversarial data undergoes processing through the BC layer, while adversarial data is excluded from further communication with the sensor. After the AI layer classifies the sensor data, only non-adversarial data is directed to the subsequent layer, namely the BC layer. This layer plays a crucial role in safeguarding the IoT sensor communication across various applications against many security threats, including injection, hijack-

ing, spoofing, etc. This information is pivotal in enabling remote patient care, empowering healthcare providers to intervene promptly and tailor treatments as needed. Moreover, it aids in the early identification of any irregularities or potential health issues, enabling a proactive healthcare network and preventive measures. The Consensus approach for mining the healthcare record in the proposed system employs the Proof of Stake (PoS) consensus mechanism, which replaces the energy-intensive Proof of Work (PoW). PoS is particularly advantageous for healthcare systems due to its energy efficiency and scalability. The PoS consensus mechanism ensures that all nodes in the network agree on the ledger's state, thereby preventing unauthorized modifications to healthcare records. This is crucial for maintaining the integrity and security of vast amounts of data generated by healthcare systems. Furthermore, PoS enhances scalability by enabling the network to handle more transactions with faster confirmation times than PoW. Reduced bottlenecks and increased transaction throughput are critical for real-time healthcare applications where timely data access and processing are essential.

4.3 Dataset description

The dataset we gathered contains recordings of body movements and vital signs from ten volunteers engaging in 12 different physical activities. We used wearable sensors on the chest, right wrist, and left ankle, secured with elastic straps. These sensors were instrumental in capturing various aspects of body motion, such as acceleration, turning speed, and magnetic field orientation. This multi-sensor approach enabled a more comprehensive measurement of body dynamics across different body parts (Figs. 11, 12 and 13).

4.4 Evaluation metrics

Evaluation metrics help us compare different models quickly to see how well they're trained and perform accurately on data. We look at various metrics to pick the best ML classifier. By checking these metrics, we can choose the model that suits our needs the most. (Table 9) describes the various evaluation metrics of the various models.

4.4.1 Random forest

The Random Forest method is an ensemble learning strategy that constructs multiple decision trees during training. Each tree in the forest is formed using a random subset of features and a random subset of training data. By consolidating predictions from various trees, Random Forest addresses the risk of overfitting in individual decision trees. It adeptly manages noise, ensuring accuracy, especially in datasets with many features and potential outliers.

4.4.2 Decision tree

A Decision Tree is a diagrammatic structure in which each internal node signifies a feature test, each branch illustrates the corresponding outcome, and each leaf node denotes the class label or a continuous value. It necessitates minimal data preparation, as it effectively handles numerical and categorical data without normalization or scaling.

4.4.3 Logistic regression

Logistic Regression is a linear model used for binary classification. It calculates the probability that a given input belongs

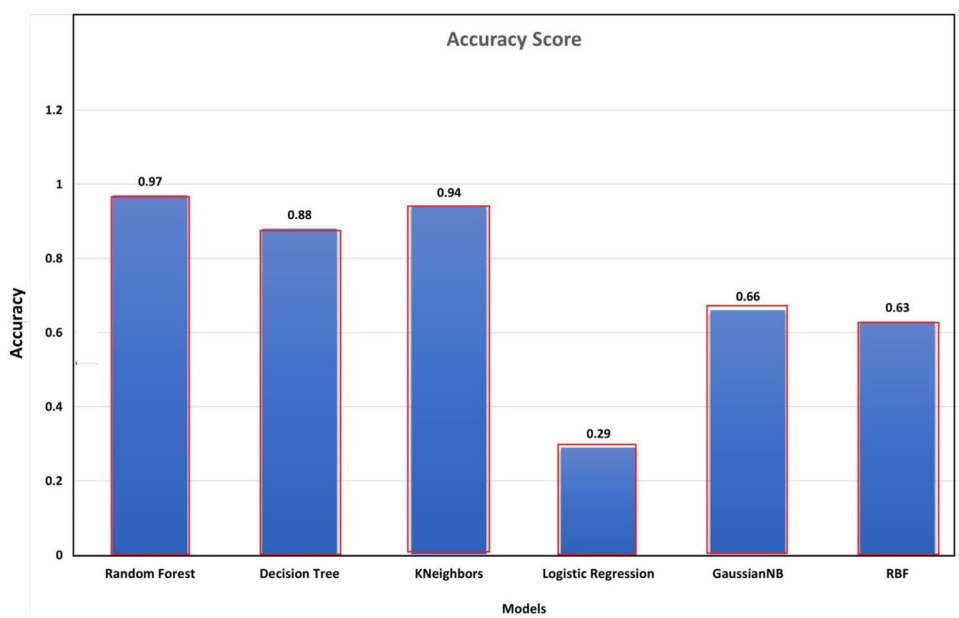
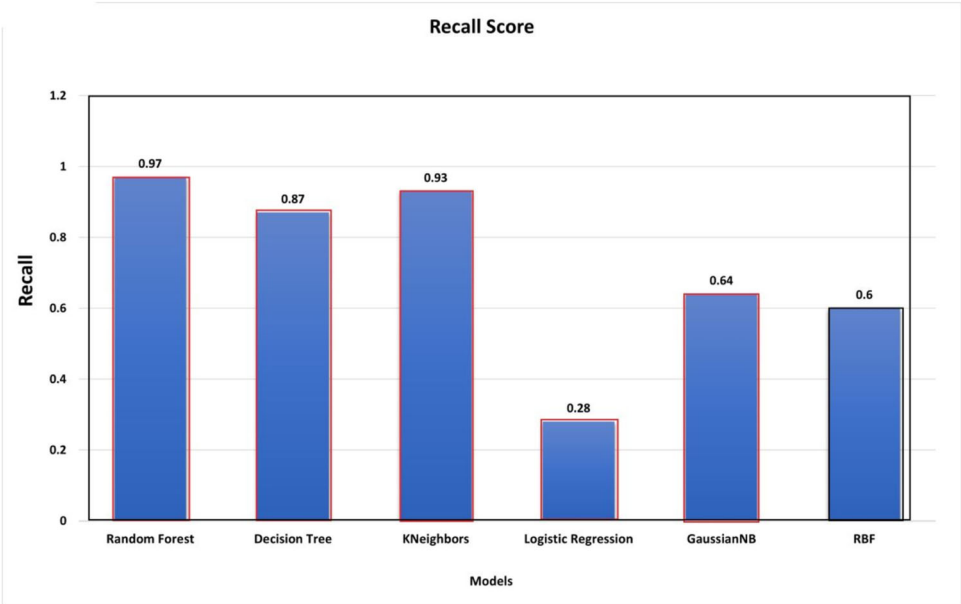


Fig. 11 Accuracy comparison

Fig. 12 Recall comparison



to a particular class and then makes predictions based on a chosen threshold.

4.4.4 Gaussian naive bayes

Gaussian Naive Bayes is a probabilistic classification algorithm based on Bayes' theorem with the assumption of independence between features. It calculates the probability of a given sample belonging to a particular class.

4.4.5 K-Nearest neighbors

K-Nearest Neighbors is a simple, instance-based learning algorithm that classifies data points based on the majority class among their nearest neighbors in the feature space. It memorizes the entire training dataset, which makes the training process very fast.

4.4.6 Support vector machine (SVM)

SVM with RBF Kernel is a powerful supervised learning algorithm used for classification, regression, and outlier detection. It constructs a hyperplane or set of hyperplanes in a high-dimensional space. It performs well even in high-dimensional spaces and is effective in cases where the data isn't linearly separable. It utilizes different kernel functions to handle various types of data distributions.

4.5 Confusion matrix

A confusion matrix serves as a visual and summarized representation of how well a classification algorithm performs. It consists of four critical components that define the evaluation metrics of the classifier:

1. True Positive (τ): Correctly classified positive labels.
2. False Positive (ϕ): Negative labels are erroneously classified as positive.
3. True Negative (υ): Correctly classified negative labels.
4. False Negative (χ): Positive labels are erroneously classified as negative.

4.6 Security and privacy

The incorporation of BC technology offers significant advancements in protecting data privacy and security within the IoT security system for healthcare. BC establishes a highly secure and unchangeable ledger using a decentralized structure and sophisticated encryption methods, enhancing the reliability of healthcare information shared across interconnected devices. This system prevents unauthorized access and alterations, giving patients greater control over their sensitive data while enabling secure sharing among authorized parties. Smart contracts within the BC automate access permissions, ensuring only approved individuals can access specific information. This reinforces the confidentiality and integrity of healthcare data in the IoT environment.

5 Future scope and research direction

It is increasingly evident that the Internet of Things (IoT) will heavily rely on Blockchain (BC) for enhanced security and efficient device networks in the coming years. Several prospective avenues for integrating Blockchain in IoT device

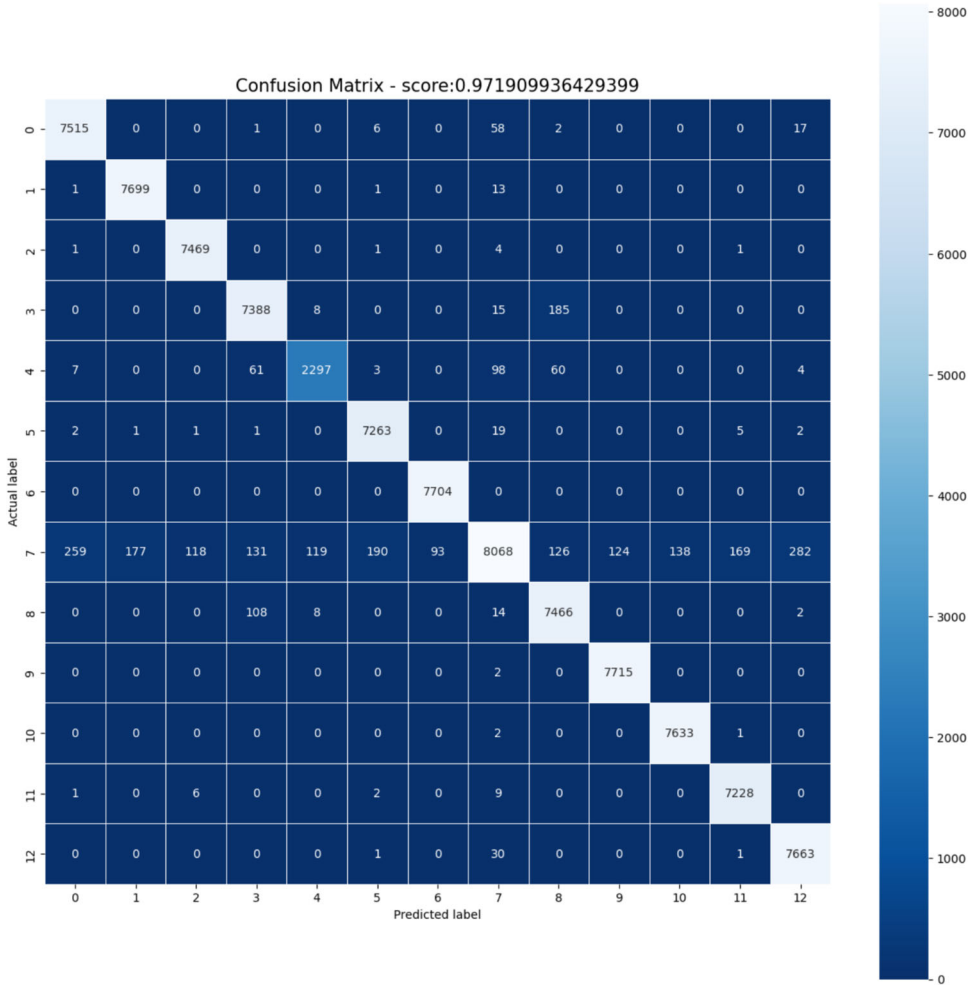


Fig. 13 Confusion Matrix of Random Forest

networks and security can be delineated: Firstly, as the IoT ecosystem continues its proliferation, an imperative concern is the fortification of security measures. Future research should be geared towards devising advanced security mechanisms uniquely tailored for blockchain-based IoT device networks. This entails exploring innovative approaches, including hardware-based security, secure enclave technology, and blockchain-specific encryption algorithms. These measures will be pivotal in safeguarding IoT devices from emerging threats. Secondly, establishing an immutable device identity is paramount in ensuring a secure IoT device network. Future directions should encompass the development of inflexible device identity solutions utilizing blockchain technology. These efforts may involve the exploration of decentralized identity network frameworks, self-sovereign identity, and verifiable credentials, all of which will guarantee the authenticity and integrity of IoT devices within the blockchain network. Moreover, the transparency and immutability of BC can revolutionize trust and reputation systems for IoT devices. Future research should focus on designing and implementing trust models that leverage these blockchain attributes to assess the behavior and reliability of

Table 9 Evaluation Metrics

Model	Accuracy	Precision	Recall	F1 Score	Sensitivity
Random Forest	97.19%	97.28%	97.07%	97.11%	97.07%
Decision Tree	87.04%	86.55%	87.03%	86.52%	87.03%
KNeighbors	93.83%	93.67%	93.57%	93.26%	93.57%
Logistic Regression	29.40%	34.25%	28.00%	24.48%	28.00%
GaussianNB	66.18%	64.38%	64.68%	62.61%	64.68%
SVM	63.33%	83.41%	60.85%	61.38%	60.85%

IoT devices. Such systems can be transformative in identifying malicious devices and fostering trustworthy interactions within the IoT ecosystem. In the face of privacy concerns arising from the vast amounts of sensitive data generated by IoT devices, future directions should be aimed at creating privacy-preserving mechanisms that facilitate secure data sharing while maintaining confidentiality. Techniques such as zero-knowledge proofs, differential privacy, and selective disclosure can be explored to enable privacy-preserving data transactions within the blockchain network. Addressing scalability is a key challenge for blockchain-based IoT device network systems. Future research should be dedicated to developing scalable consensus mechanisms specifically designed for IoT environments. This includes an exploration of consensus algorithms like proof-of-stake (PoS), delegated proof-of-stake (DPoS), and practical Byzantine fault tolerance (PBFT) variants, all of which offer high throughput and low latency, thereby optimizing the efficiency of the IoT device network. To achieve seamless integration, interoperability among blockchain networks and IoT devices is imperative. Future directions should encompass the formulation of standardized protocols and frameworks that facilitate interoperability and communication between diverse blockchain-based IoT device network systems. These standards foster collaboration and promote secure data exchange across varied IoT ecosystems. Energy efficiency is a pivotal consideration for IoT devices with limited power sources. Subsequent research should be aimed at optimizing energy consumption in blockchain-based IoT device network systems [148]. Techniques such as energy-aware consensus algorithms, dynamic resource allocation, and energy-efficient data storage mechanisms can be explored to maximize the operational lifespan of IoT devices while harnessing the potential of blockchain technology. As the adoption of blockchain-based IoT device network systems expands, it becomes increasingly necessary to establish clear regulatory and governance frameworks. Future directions should encompass the development of regulatory guidelines and governance models to ensure compliance, transparency, and accountability in the deployment and operation of these systems. This will engender trust and responsible utilization of the technology [149]. Lastly, in pursuit of a robust and secure blockchain-based IoT ecosystem, future research should diligently identify and address potential vulnerabilities and threats. This includes exploring mitigation strategies for threats such as 51% attacks, Sybil attacks, and smart contract vulnerabilities. By fortifying security measures and conducting comprehensive risk assessments, the resilience of blockchain-based IoT device network systems can be significantly enhanced. In embracing these forward-looking directions, blockchain technology can be harnessed to its fullest potential, facilitating secure, scalable, and privacy-preserving solutions for the rapidly burgeoning IoT landscape.

6 Conclusions

This paper comprehensively reviews existing literature, focusing on integrating BC and the IoT as the primary components to enhance research solutions for bolstering the security of IoT devices. Blockchain offers a decentralized, transparent, and secure infrastructure, providing opportunities to improve security, data integrity, and trust within IoT ecosystems. Incumbent IoT systems encounter challenges such as heterogeneity, poor interoperability, resource constraints, and privacy and security vulnerabilities. The recent emergence of blockchain technologies is a promising solution, addressing enhanced interoperability, privacy, security, traceability, and reliability issues. This paper explores blockchain-driven security for IoT networks, aiming to deliver tamper-resistant, traceable, and heightened security measures. A Smart healthcare case study also illustrates secure communication among IoT devices within healthcare settings. Our proposed system outlines a methodology for sharing healthcare data through utilizing Artificial Intelligence (AI)/ML and Smart Contracts, capitalizing on the advantages of IPFS. The proposed model offers enhanced security by leveraging blockchain technology, reducing single points of failure and improving data integrity with immutable records and smart contracts. Data privacy is enhanced through ML algorithms encrypt patient data and manage access control. Blockchain standardizes data formats, facilitating interoperability and enabling real-time, secure data sharing. The model's use of distributed processing and automation improves efficiency and scalability. However, integrating ML and blockchain is complex and costly, with performance issues such as limited transaction throughput and high latency. Additionally, the resource-intensive nature of these technologies increases energy consumption and costs, and privacy and regulatory challenges persist.

Author Contributions VM wrote the main manuscript, VR conceptualized the manuscript, reviewed and edited it, MY worked on the creation of tables, and identification of challenges, MS worked on figures and their detailing, PY worked on the coherence among various sections/modules of the manuscript, UA worked on the proposed system, case study and the validation part, RC worked with overall language, editing and revisions.

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Data Availability Although the data is confidential and still in use for further extension of the work, it can be furnished based on a genuine request.

Declarations

Competing interests The authors declare no competing interests.

Ethics Approval Not Applicable

Consent to publish Authors have no conflict, and they have consent to publish.

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